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ATG6 interacting with NPR1 increases *Arabidopsis thaliana* resistance to *Pst* DC3000/*avrRps4* by increasing its nuclear accumulation and stability

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Abstract

Autophagy-related gene 6 (ATG6) plays a crucial role in plant immunity. Nonexpressor of pathogenesis-related genes1 (NPR1) acts as a signaling hub of plant immunity. However, the relationship between ATG6 and NPR1 is unclear. Here, we find that ATG6 directly interacts with NPR1. *ATG6* overexpression significantly increased nuclear accumulation of NPR1. Furthermore, we demonstrate that *ATG6* increases NPR1 protein levels and improves its stability. Interestingly, ATG6 promotes the formation of SINCs (SA-induced NPR1 condensates)-like condensates. Additionally, ATG6 and NPR1 synergistically promote the expression of *pathogenesis-related* genes. Further results showed that silencing *ATG6* in *NPR1-GFP* exacerbates *Pst* DC3000/*avrRps4* invasion, while double overexpression of *ATG6* and *NPR1* synergistically inhibits *Pst* DC3000/*avrRps4* invasion. In summary, our findings unveil an interplay of NPR1 with ATG6 and elucidate important molecular mechanisms for enhancing plant immunity.

Highlight

We unveil a novel relationship in which ATG6 positively regulates NPR1 in plant immunity.

eLife Assessment

This study, which proposes a new role of ATG6 in plant immune response, makes a **valuable** contribution to our understanding of plant immunity. The results suggest a direct interaction between ATG6 and NPR1, a salicylic acid receptor protein, and they will be of interest to scientists studying the regulation of plant immunity. The data presented are **convincing**, although the discrepancies between data from fluorescence microscopy and protein blots, particularly in the interpretation of ATG6-mCherry fusion proteins. Addressing these inconsistencies would enhance the study's overall impact.

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Introduction

Plants are constantly challenged by pathogens in nature. In order to survive and reproduce, plants have evolved complex mechanisms to cope with attack by pathogens (Jones and Dangl, 2006). Nonexpressor of pathogenesis- related genes 1 (NPR1) is a key regulator of plant immunity (Chen *et al*., 2021b). It contains the BTB/POZ (Broad Compex, Tramtrack, and BricaBrac/Pox virus and Zinc finger) domain in the N-terminal region, the ANK (Ankyrin repeats) domain in the middle region, and SA-binding domain (SBD) and the nuclear localization sequence (NLS) in the C-terminal region (Cao *et al*., 1997; Rochon *et al*., 2006; Kumar *et al*., 2022). NPR1 is a receptor of SA (salicylic acid) mainly localized as an oligomer in the cytoplasm and sensitive to the surrounding redox state (Tada *et al*., 2008; Wu *et al*., 2012). SA mediates the dynamic oligomer to dimer response of NPR1 (Tada *et al*., 2008) and promotes translocation of NPR1 into the nucleus, which increases plant resistance to pathogens by activating the expression of immune-related genes (Kinkema *et al*., 2000; Chen *et al*., 2021b).

NPR1 is mainly degraded by the ubiquitin proteasome system (UPS) (Spoel *et al*., 2009; Saleh *et al*., 2015; Skelly *et al*., 2019). An increasing researches have shown that autophagy and the UPS pathway play overlapping roles in regulating intracellular protein homeostasis (Zhou *et al*., 2014; Marshall *et al*., 2015; Kikuchi *et al*., 2020). Our previous study showed that ATGs (autophagy-related genes) are involved in NPR1 turnover (Gong *et al*., 2020). Autophagy negatively regulates *Pst* DC3000/*avrRpm1*-induced programmed cell death (PCD) via the SA receptor NPR1 (Yoshimoto *et al*., 2009). These results imply that ATGs might be involved in plant immunity through the regulation of NPR1 homeostasis. However, the detailed mechanism has not yet been elucidated.

ATG6 is the homologues of yeast Vps30/Atg6 and mammalian BECN1/Beclin1 (Xu *et al*., 2017). It is a common and required subunit of the class III phosphatidylinositol 3-kinase (PtdIns3K) lipid kinase complexes, which regulates autophagosome nucleation in *Arabidopsis thaliana* (*Arabidopsis*) (Qi *et al*., 2017; Wang *et al*., 2020). The homozygous *atg6* mutant is lethal, suggesting that ATG6 is essential for plant growth and development (Fujiki *et al*., 2007; Qin *et al*., 2007; Harrison-Lowe and Olsen, 2008; Patel and Dinesh-Kumar, 2008). In *Arabidopsis*, *N. benthamiana* and wheat, ATG6 or its homologues was reported to act as a positive regulator to enhance plant disease resistance to *P. syringae pv. tomato* (*Pst*) DC3000 and *Pst* DC3000/*avrRpm1* bacteria (Patel and Dinesh-Kumar, 2008), *N. benthamiana* mosaic virus (TMV) (Liu *et al*., 2005), turnip mosaic virus (TuMV) (Li *et al*., 2018), pepper mild mottle virus (PMMoV) (Jiao *et al*., 2020), and *Blumeria graminis f. sp. tritici (Bgt)* fungus (Yue *et al*., 2015). Several research teams have also elucidated that ATG6 interacted with Bax Inhibitor-1 (NbBI-1) (Xu *et al*., 2017) and RNA-dependent RNA polymerase (RdRp) (Li *et al*., 2018) to suppress pathogen invasion. However, the mechanism by which ATG6 suppresses pathogen invasion by regulating NPR1 has not yet been reported.

Here, we show that ATG6 and NPR1 synergistically enhance *Arabidopsis* resistance to *Pst* DC3000/*avrRps4* infiltration. We discover that ATG6 increases NPR1 protein levels and nuclear accumulation of NPR1. Moreover, ATG6 can stabilize NPR1 and promote the formation of SINCs (SA-induced NPR1 condensates)-like condensates. Our study revealed a unique mechanism in which NPR1 cooperatively increases plant immunity with ATG6.

Results

NPR1 physically interacts with ATG6 in vitro and in vivo

To examine the relationship between ATGs and NPRs, we predicted that some ATGs might interact with NPRs. In a yeast two-hybrid (Y2H) screen, we identified that NPR1, NPR3 and NPR4 could interact with ATG6 and several ATG8 isoforms (**Fig. S1** and **Results S1** in the Supplemental Data 1). In this study, we mainly investigated the relationship between ATG6 and NPR1 during the process of plant immune response. Firstly, the NPR1 truncations NPR1-N (1-328AA, containing the BTB/POZ domain, ANK1, ANK2,) and NPR1-C (328-594AA, containing the ANK3, ANK4, SA-binding domain (SBD) and NLS) were used to identify the interaction domains between NPR1 and ATG6. The results showed that NPR1-C interacted with full-length ATG6 in yeast (**Fig. 1a**, line 3). The interaction between NPR1 and SnRK2.8 was used as a positive control (Lee *et al*., 2015). Secondly, pull-down assays were performed *in vitro*. NPR1-His bound GST-ATG6, but not GST (**Fig. 1b**). Furthermore, co-immunoprecipitation (Co-IP) assays were performed in *N. benthamiana*, as shown in **Fig. 1c**, ATG6-mCherry was co- immunoprecipitated with NPR1-GFP. In **Fig. S2**, fluorescence signals of NPR1-GFP and ATG6-mCherry were co-localized in the nucleus under normal and 1 mM SA treatment conditions. The interaction between ATG6 and NPR1 was also verified by a bimolecular fluorescence complementation (BiFC) assay (**Fig. 1d** and **e**). These results demonstrate that ATG6 interacts with NPR1 both *in vitro* and *in vivo*.

ATG6 co-localized with NPR1 in the nucleus

Remarkably, under normal and SA treatment conditions, we found that ATG6 is localized in the cytoplasm and nucleus, and it co-localized with NPR1 in the nucleus (**Fig. S2**). Nuclear localization of ATG6 was also observed in *N. benthamiana* transiently transformed with ATG6-mCherry and ATG6-GFP under normal and SA treatment conditions (**Fig. 2a** and **b**). ATG6-GFP co- localizes with the nuclear localization marker nls-mCherry (indicated by white arrows) (**Fig. 2b**). Additionally, we observed punctate patterns indicative of canonical autophagy-like localization of ATG6-GFP fluorescence signals (indicated by red circles) (**Fig. 2b**). The nuclear localization signal of ATG6 was also observed in *UBQ10::ATG6-GFP* overexpressing *Arabidopsis* (**Fig. S3a**). To exclude the possibility that the observed localization of ATG6-GFP is due to free GFP. The protein levels of ATG6-GFP and free GFP in *UBQ10::ATG6-GFP Arabidopsis* and *N.benthamiana* were detected before and after SA treatment. Notably, no free GFP was detected and this means that the fluorescence signal observed by confocal microscopy is ATG6-GFP, not free GFP (**Fig. S3d** and **e**). Furthermore, we analyzed the putative nuclear localization signal (NLS) in the ATG6 protein sequence using the online INSP (Identification of Nuclear Signal Peptide) prediction software (*http://www.csbio.sjtu.edu.cn/bioinf/INSP/*). The prediction results indicated the presence of a potential nuclear localization sequence "FLKEKKKKK" within the ATG6 protein, spanning from the 217th to the 223rd amino acid (**Fig. 2c**). We also utilized INSP to investigate the NLS of various ATG proteins (TaATG6a (Yue *et al*., 2015), TaATG6b (Yue *et al*., 2015), TaATG6c (Yue *et al*., 2015),

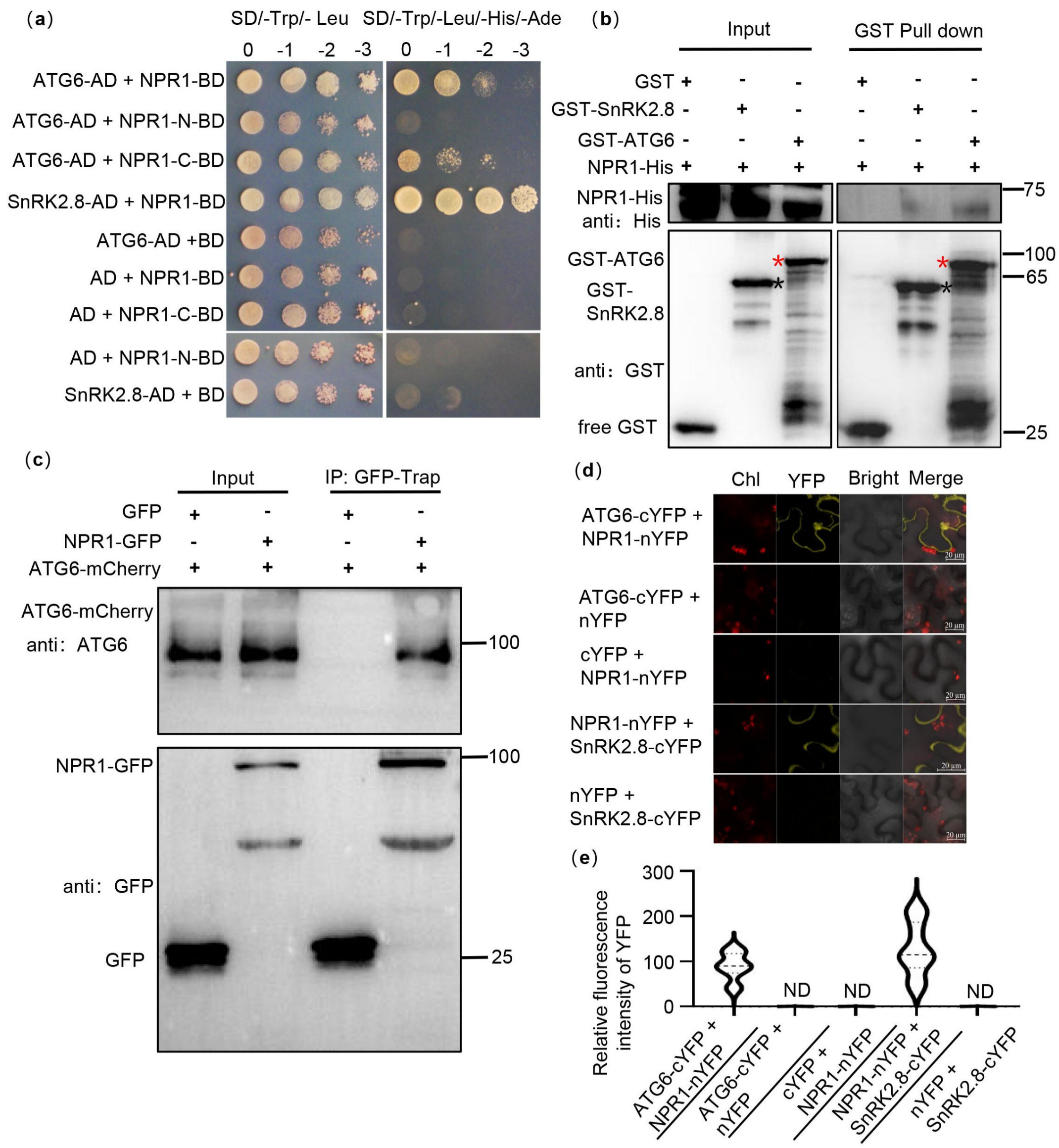

Figure 1.

Physical interaction between NPR1 and ATG6.

(a). Interaction of NPR1 with ATG6 in yeast. The CDS of *ATG6, NPR1, NPR1-N* (0-328AA) and *NPR1-C* (328-594AA), *SnRK2.8* were fused to pGADT? (AD) and pGBKT? (BO), respectively. Co-transformation of NPR1-BD + AD, BO+ ATG6-AD, BO+ SnRK2.8-AD, NPR1-N-BD + AD, NPR1-C-BD + AD were used as negative controls. The interaction of NPR1-BD and SnRK2.8-AD was used as a positive control. Yeast growth on SD/-Trp-Leu-His-Ade media represents interaction. Numbers represent the dilution fold of yeast. 0, -1 (10 fold dilution), -2 (100 fold dilution) and -3 (1000 fold dilution). **(b).** In vitro pull-down assays of NPR1-His with GST-ATG6 fusion protein. NPR1-His prokaryotic proteins were incubated with GST-tag Purification Resin conjugated with GST-ATG6, GST and GST-SnRK2.8. Western blotting analysis with anti-GST and anti-His. Black asterisk(*) indicate GST-SnRK2.8 bands. Red asterisk(*) indicate GST-ATG6 bands. **(c).** Co-immunoprecipitation of NPR1 with ATG6 in *vivo.* Total protein was extracted from *N. benthamiana* transiently transformed with ATG6-mCherry + GFP and ATG6- mCherry + NPR1-GFP, followed by IP with GFP-Trap. Western blots analysis with ATG6 and GFP antibodies. **(d).** Bimolecular fluorescence complementation assay of NPR1 with ATG6 in *Nicotiana benthamiana* leaves. Agrobacterium carrying ATG6-nYFP and NPR1-cYFP was co-expressed in leaves of *Nicotiana benthamiana* for 3 days. As a positive control, NPR1-nYFP and SnRK2.8-cYFP were co-expressed. As negative controls, nYFP and ATG6-cYFP, NPR1-nYFP and cYFP, nYFP and SnRK2.8-cYFP were co-expressed. Confocal images were obtained from chloroplast (Chi), YFP, Bright field. Scale bar= 20 μm. (e). Relative fluorescence intensity of YFP in **(d)** using image J software, ND means not detected, n = 15 independent images were analyzed to quantify YFP fluorescence. All experiments were performed with three biological replicates.

SlATG8h (Li *et al*., 2020)) that have been previously reported to localize in the nucleus. This analysis revealed a relatively conserved NLS sequence motif: "E/K-K/E-K-K-L/K-K" in these ATG proteins (**Fig. 2c**).

Moreover, the nuclear and cytoplasmic fractions were separated. Under SA treatment, ATG6-mCherry and ATG6-GFP were detected in the cytoplasmic and nuclear fractions in *N. benthamiana* (**Fig. 2d and e**). However, in *N. benthamiana*, we observed that ATG6-mCherry was not detected in the nuclear fractions under normal conditions, which differents with the results shown in **Fig. 2a**. We suspect that this discrepancy may be due to the fluorescence signal in **Fig. 2a** primarily arising from free mCherry rather than the ATG6-mCherry fusion. ATG6 was also detected in the nuclear fraction of *UBQ10::ATG6-GFP* and *UBQ10::ATG6-mCherry* overexpressing plants, and SA promoted both cytoplasm and nuclear accumulation of ATG6 (**Fig. 2f and g**). Additionally, we obtained *ATG6* and *NPR1* double overexpression of *Arabidopsis UBQ10::ATG6-mCherry* × *35S::NPR1-GFP* (*ATG6-mCherry* ×*NPR1-GFP*) by crossing and screening (**Fig. S4a**). In *ATG6- mCherry* × *NPR1-GFP*, we observed co-localization of ATG6-mCherry with NPR1-GFP in the nucleus (**Fig. S3b**). These results are consistent with the prediction of the subcellular location of ATG6 in the *Arabidopsis* subcellular database (*https://suba.live/*) (**Fig. S3c**). Additionally, we have conducted an investigation into the localization of endogenous ATG6 in Col. Our results demonstrate that endogenous ATG6 is present in both the nucleus and cytoplasm, and we have observed that SA treatment promotes the accumulation of ATG6 in the nucleus (**Fig. S5**). Together, these findings suggest that ATG6 is localized to both cytoplasm and nucleus, and co- localized with NPR1 in the nucleus.

ATG6 overexpression increased nuclear accumulation of NPR1

Previous studies have shown that the nuclear localization of NPR1 is essential for improving plant immunity (Kinkema *et al*., 2000; Chen *et al*., 2021b). We observed that a stronger nuclear localization signal of NPR1-GFP in *ATG6-mCherry* × *NPR1-GFP* leaves than that in *NPR1-GFP* under normal condition and 0.5 mM SA treatment for 3 h (**Fig. 3a-b** and **Fig. S6**). These findings indicate that ATG6 might increase nuclear accumulation of NPR1. To exclude the possibility that the observed localization of NPR1-GFP is due to free GFP, we detected the levels of NPR1-GFP and free GFP in *ATG6- mCherry* x *NPR1-GFP* plants before and after SA treatment. Only ∼ 10 % of free GFP was detected in *ATG6-mCherry* x *NPR1-GFP* plants before and after SA treatment, confirming that the observed localization of NPR1-GFP is not due to free GFP (**Fig. S4b**). Furthermore, the nuclear and cytoplasmic fractions of *ATG6-mCherry* × *NPR1-GFP* and *NPR1-GFP* were separated. Under normal conditions, the nuclear fractions NPR1-GFP in *ATG6-mCherry* × *NPR1-GFP* and *NPR1-GFP* were relatively weaker (**Fig. 3c**), which differs from the above observation (**Fig. 3a**). We speculate that this phenomenon might be attributed to the rapid turnover of NPR1 in the nucleus (Spoel *et al*., 2009; Saleh *et al*., 2015). Consistent with the fluorescence distribution results, the nuclear fractions of NPR1-GFP in *ATG6-mCherry* × *NPR1-GFP* were significantly higher than those in *NPR1-GFP* under 0.5 mM SA treatment for 3 and 6 h (**Fig. 3c** and **Fig. S7**). Furthermore, *Agrobacterium* harboring ATG6- mCherry and NPR1-GFP were transiently transformed to *N. benthamiana* leaves. After 1 d, the leaves were treated with 1 mM SA for 8 and 20 h.

Subsequently nucleoplasmic separation experiments were performed. Similar to *Arabidopsis*, increased nuclear accumulation of NPR1 was found when *ATG6* was overexpressed (**Fig. 3e** and **Fig. S7**). Notably, we found that the ratio (nucleus NPR1/total NPR1) in *ATG6-mCherry* × *NPR1-GFP* was not significantly different from that in *NPR1-GFP* after SA treatment, and a similar phenomenon was observed in *N. benthamiana* (**Fig. 3d**, **3f** and **Fig. S7**). These results suggested that the increased nuclear accumulation of NPR1 in *ATG6-mCherry* x *NPR1-GFP* plants might attributed to higher levels and more stable NPR1 rather than the enhanced nuclear translocation of NPR1 facilitated by ATG6. Furthermore, we validated the functionality of the ATG6- GFP and ATG6-mCherry fusion proteins utilized in this study by examining the phenotypes of *ATG6-GFP* and *ATG6-mCherry Arabidopsis* plants under carbon starvation conditions (**Fig. S8** and **Results S2** in the Supplemental Data 1).

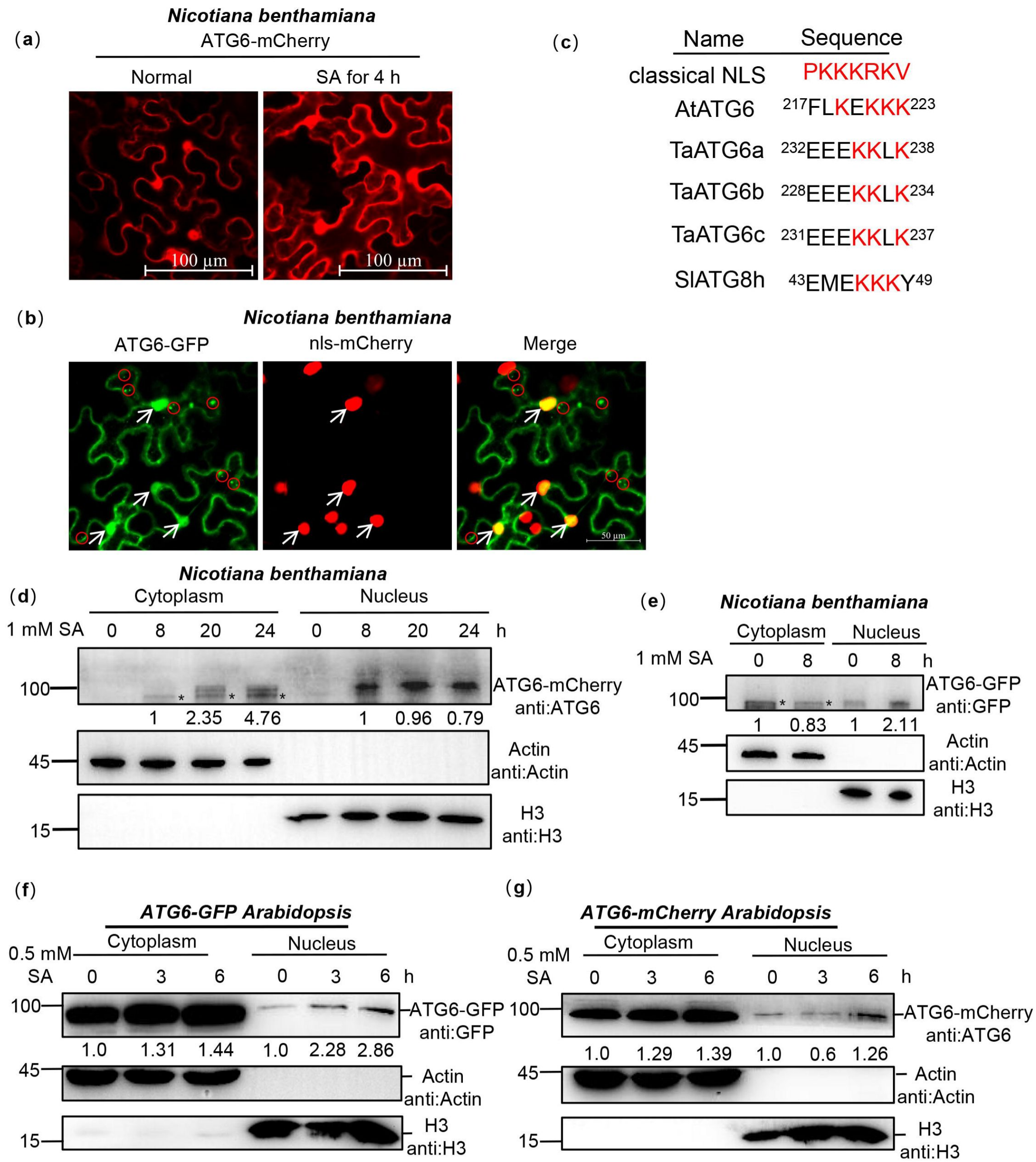

Figure 2.

ATG6 is localized in the cytoplasm and nucleus.

(a). The nuclear localization of ATG6-mCherry in *N. benthamiana.* Scale bar, 100 μm. **(b).** Co-localization of ATG6-GFP and nls-mCherry in *N. benthamiana.* Scale bar, 50 μm. **(c).** ATGs protein nuclear localization sequence analysis using the online INSP (Identification of Nuclear Signal Peptide) prediction software. **(d).** Subcellular fractionation of ATG6-mCherry in *N. benthamiana* after 1 mM SA treatment. Black asterisk(*) indicate ATG6-mCherry bands. **(e).** Subcellular fractionation of ATG6-GFP in *N. benthamiana* after 1 mM SA treatment. Black asterisk(*) indicate ATG6-GFP bands. **(f).** Subcellular fractionation of ATG6-GFP in *ATG6-GFP Arabidopsis* after 0.5 mM SA treatment. **(g).** Subcellular fractionation of ATG6-mCherry in *ATG6-mCherry Arabidopsis* after 0.5 mM SA treatment. In **(d-g),** ATG6-mCherry **(d** and **g)** and ATG6-GFP **(e** and **f)** were detected using ATG6 or GFP antibody. Actin and H3 were used as cytoplasmic and nucleus internal reference, respectively. Numerical values indicate quantitative analysis of ATG6-mCherry and

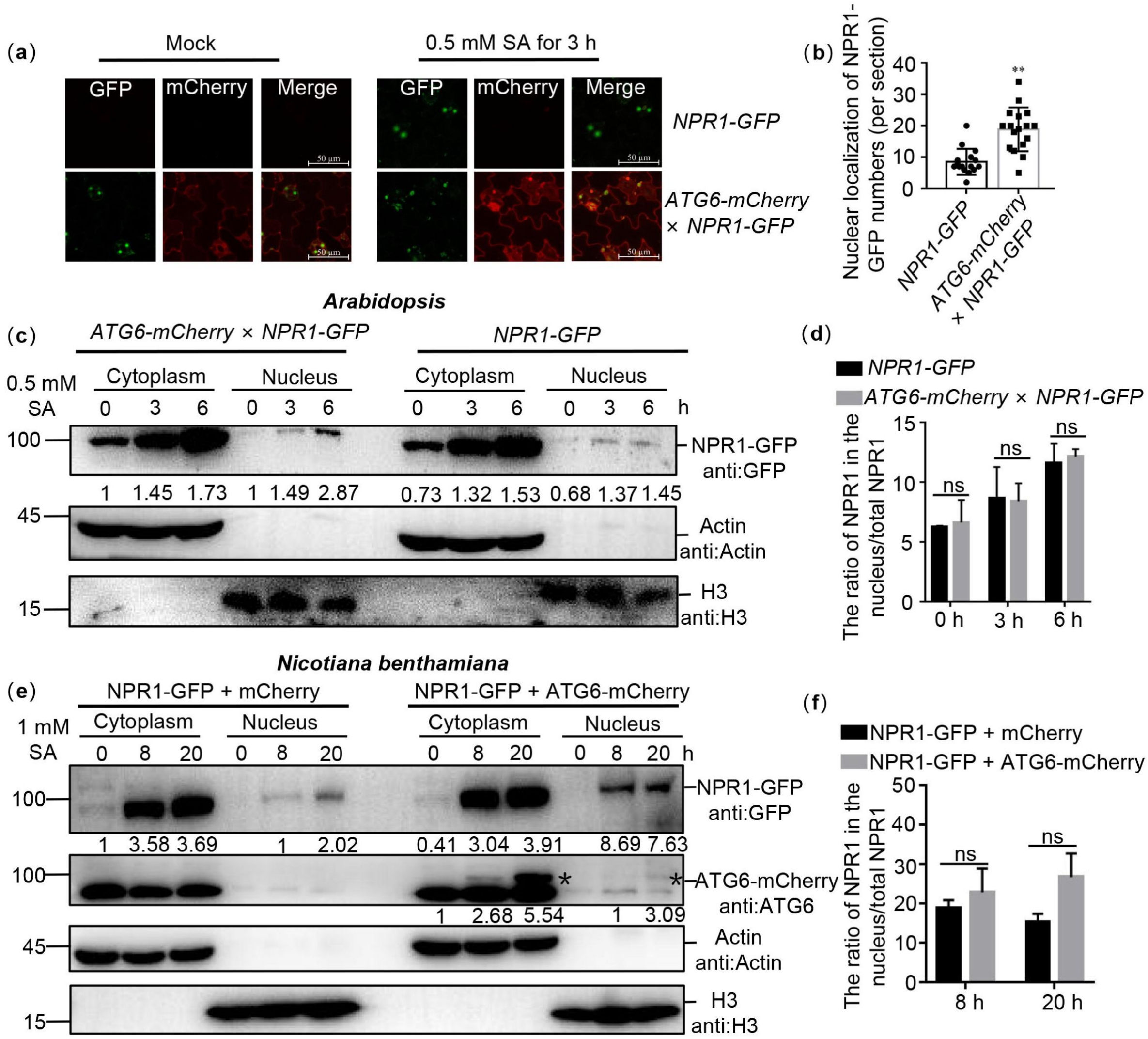

Figure 3.

ATG6 increases the nuclear accumulation of NPR1 under SA treatment.

(a). Confocal images of NPR1-GFP nuclear localization in 7-day-old seedlings of *NPR1-GFP* and *ATG6-mCherry* x *NPR1-GFP* under normal and 0.5 mM SA spray for 3 h. Scale bar, 50 µm. **(b).** The count of nuclear localizations of NPR1-GFP in *ATG6-mCherry* x *NPR1-GFP* and *NPR1-GFP Arabidopsis* plants following SA treatment. in **(a),** Statistical data were obtained from three independent experiments, each comprising five individual images, resulting in a total of 15 images analyzed for this comparison. ** indicates that the significant difference compared to the control is at the level of 0.01 (Student t test p value,** p< 0.01). **(c).** Subcellular fractionation of NPR1-GFP in 7-day-old seedlings of *NPR1-GFP* and *ATG6-mCherry* x *NPR1-GFP* after 0.5 mM SA treatment for 0, 3 and 6 h. **(d).** The ration of NPR1 in the nucleus/total NPR1 in **(c),** ns indicates no significant difference. **(e).** Subcellular fractionation of NPR1-GFP in *N. benthamiana* after 1 mM SA treatment for 0, 8 and 20 h. Black asterisk(*) indicate ATG6-mCherry bands. **(f).** The ration of NPR1 in the nucleus/total NPR1 in **(e),** ns indicates no significant difference. In **(c** and **e),** cytoplasmic and nuclear proteins were extracted from *Arabidopsis* or *N. benthamiana.* NPR1-GFP were detected using GFP antibody. Actin and H3 were used as cytoplasmic and nucleus internal reference, respectively. Numerical values indicate quantitative analysis of NPR1-GFP using

ATG6 increases endogenous SA levels and promotes the expression of NPR1 downstream target genes

NPR1 localized in the nucleus is essential for activation of immune gene expression (Kinkema *et al.*, 2000; Chen *et al.*, 2021b). In our study, we observed that *ATG6* overexpression increased nuclear accumulation of NPR1 (**Fig. 3**) and demonstrated an interaction between ATG6 and NPR1 in the nucleus (**Fig. 1d**). Therefore, we speculate that ATG6 might regulate NPR1 transcriptional activity. Notably, the expression level of *ICS1* in *ATG6-mCherry* × *NPR1-GFP* seedlings was significantly higher than that in *NPR1-GFP* under normal and SA treatment conditions (**Fig. S9**). Free SA levels in *ATG6- mCherry* × *NPR1-GFP* were also significantly higher compared to *NPR1-GFP* under *Pst* DC3000/*avrRps4* treatment. While there was no significant difference was observed under normal condition (**Fig. 4a**), this may be related to free SA consumption, as it can be converted to bound SA (Ding and Ding, 2020). In addition, the expression of *PR1* (*pathogenesis-related* gene) and *PR5* in *ATG6-mCherry* × *NPR1-GFP* was significantly higher than that of *NPR1-GFP* under normal and SA treatment conditions (**Fig. 4b and c**). The expression of *PR1* and *PR5* in *ATG6-mCherry* was significantly higher than that of Col under *Pst* DC3000/*avrRps4* treatment (**Fig. S10**). These results support the role of ATG6 in facilitating the expression of NPR1 downstream *PR1* and *PR5* genes.

ATG6 increases NPR1 protein levels and the formation of SINCs-like condensates

Interestingly, similar to previous reports (Zavaliev *et al.*, 2020), SA promoted the translocation of NPR1 into the nucleus, but still a significant amount of NPR1 was present in the cytoplasm (**Fig. 3c and e**). Previous studies have shown that SA increased NPR1 protein levels and facilitated the formation of SINCs in the cytoplasm, which are known to promote cell survival (Zavaliev *et al.*, 2020). In our experiments, we observed that under SA treatment, the protein levels of NPR1 in *ATG6-mCherry* × *NPR1-GFP* was significantly higher than that in *NPR1-GFP* (**Fig. 5a**). To further support our conclusions, we proceeded to silence *ATG6* in *NPR1-GFP* (*NPR1-GFP*/silencing *ATG6*) and subsequently assessed the protein level of NPR1-GFP before and after SA treatment. Our findings revealed that the protein level of NPR1-GFP in *NPR1-GFP*/silencing *ATG6* under SA treatment was notably lower than that in the *NPR1-GFP*/Negative control **(Fig. S11)**. Under SA treatment for 8 h, the protein levels of NPR1-GFP in *N. benthamiana* co-transformed with ATG6-mCherry + NPR1-GFP was also significantly higher than that of mCherry + NPR1-GFP (**Fig. 5b**). While there was a slight increase at 20 h, a minor decrease was observed at 24 h, suggesting that the rise in NPR1 protein levels induced by ATG6 was transient. We also detected the expression of *NPR1* was detected. It is worth noting that NPR1 up-regulation was more obvious in Col after 3 h treatment with *Pst* DC3000/*avrRps4*. After 6 h treatment with *Pst* DC3000/*avrRps4*, there was no significant difference in the expression of NPR1 between Col and *ATG6-mCherry* (**Fig. S12**). These results suggest that ATG6 increases NPR1 protein levels. After SA treatment, more SINCs-like condensates fluorescence were observed in *N. benthamiana* co-transformed with ATG6-mCherry + NPR1-GFP compared to mCherry + NPR1-GFP (**Fig. 5c-d and Supplemental movie 1-2**). Additionally, we observed that SINCs-like condensates signaling partial co-localized with certain ATG6-mCherry autophagosomes fluorescence signals (**Fig. S13**). Taken together, these results suggest that ATG6 increases the protein levels of NPR1 and promotes the formation of SINCs-like condensates, possibly caused by ATG6 increasing SA levels *in vivo*.

ATG6 maintains the protein stability of NPR1

Maintaining the stability of NPR1 is critical for enhancing plant immunity (Skelly *et al.*, 2019). To further verify whether ATG6 regulates NPR1 stability, we co-transfected NPR1-GFP with ATG6-mCherry or mCherry in *N. benthamiana* and performed cell-free degradation assays. Our results

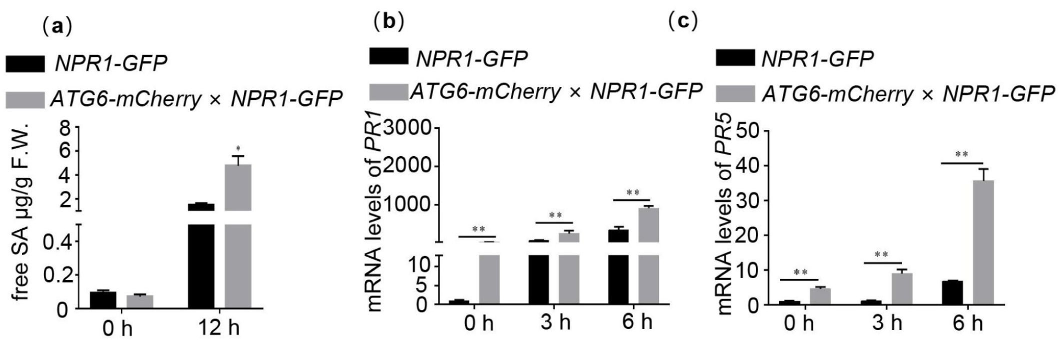

Figure 4.

ATG6 increases endogenous SA levels and promotes the expression of NPR1 downstream target genes.

(a). Level of free SA in 3-week-old *NPR1-GFP* and *ATG6-mCherry* x *NPR1-GFP* after *Pst DC3000/avrRps4* for 12 h.(b-c). Expression of *PR1* (b) and *PR5* (c) in 3-week old *NPR1-GFP* and *ATG6-mCherry* x *NPR1-GFP* under normal and SA treatment conditions, values are means ± SD (n = 3 biological replicates). The *AtActin* gene was used as the internal control. * or ** indicates that the significant difference compared to the control is at the level of 0.05 or 0.01 (Student t test p value,* p< 0.05 or** p< 0.01). All experiments were performed with three biological replicates.

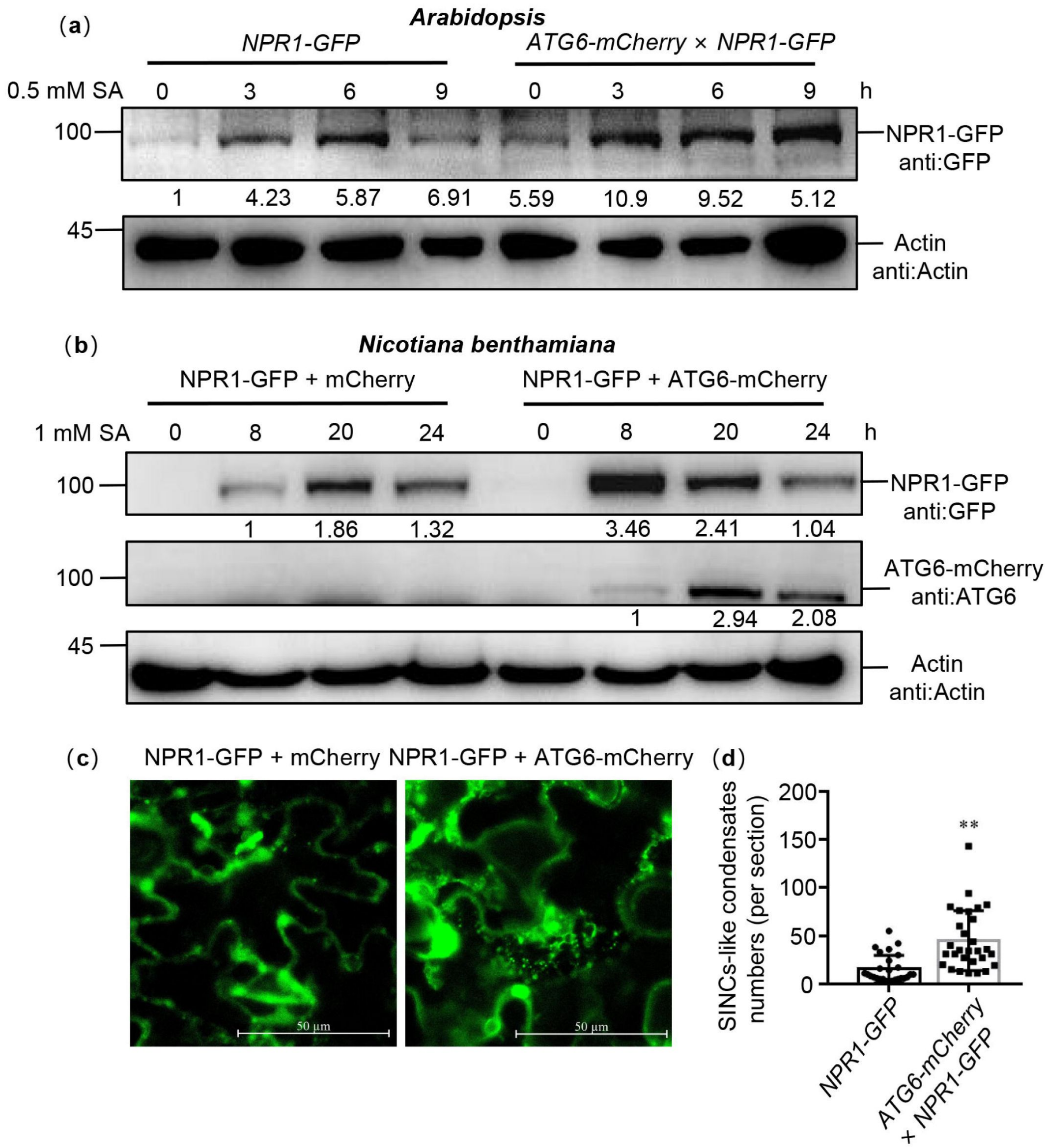

Figure 5.

ATG6 increases the NPR1 protein levels and the formation of SINCs-like condensates.

(a). NPR1-GFP protein levels in 7-day-old seedlings of *NPR1-GFP* and *ATG6- mCherry* x *NPR1-GFP* after 0.5 mM SA treatment for 0, 3, 6 and 9 h. Numerical values indicate quantitative analysis of NPR1-GFP protein using image J. **(b).** NPR1-GFP protein levels in *N. benthamiana.* ATG6-mCherry + NPR1-GFP, NPR1-GFP + mCherry were co-expressed in *N. benthamiana.* After 2 days, leaves were treated with 1 mM SA for 8, 20, 24 h. Total proteins were extracted and analyzed. Numerical values indicate quantitative analysis of NPR1-GFP protein using image J. **(c).** ATG6 promotes the formation of SINCs-like condensates. ATG6-mCherry + NPR1- GFP, NPR1-GFP + mCherry were co-expressed in *N. benthamiana.* After 2 days, leaves were treated with 1 mM SA for 24 h. Confocal images obtained at excitation with wavelengths of 488 nm, scale bar= 50 μm. **(d).** SINCs-like condensates numbers of per section in **(c),** n > 10 sections. ** indicates that the significant difference compared to the control is at the level of 0.01 (Student t test p value,** p< 0.01). All experiments were performed with three biological replicates.

showed that NPR1-GFP degradation was significantly delayed when *ATG6* was overexpressed (**Fig. S14**). A similar trend was observed in *Arabidopsis*, where the NPR1-GFP protein in *ATG6-mCherry* × *NPR1-GFP* showed a slower degradation rate compared to *NPR1-GFP* during 0~180 min time period in a cell-free degradation assay (**Fig. 6a and b**). Moreover, when *Arabidopsis* seedlings were treated with cycloheximide (CHX) to block protein synthesis, we found that NPR1-GFP in *NPR1-GFP* was degraded after CHX treatment for 3~9 h and the half-life of NPR1-GFP is ~3 h, while the half-life of NPR1- GFP in *ATG6-mCherry* × *NPR1-GFP* is ~9 h (**Fig. 6c and d**). In addition, we also analyzed the degradation of NPR1-GFP in *NPR1-GFP* and *NPR1- GFP/atg5* following 100 μM cycloheximide (CHX) treatment. The results show that the degradation rate of NPR1-GFP in *NPR1-GFP/atg5* plants was similarly to that in *NPR1-GFP* plants (**Fig. 6e and f**). These results indicate that ATG6 plays a role in maintaining the stability of NPR1, which may also be related to the fact that ATG6 promotes an increase in free SA *in vivo*, since SA has the function of increasing NPR1 stability (Ding *et al*., 2016; Skelly *et al*., 2019).

ATG6 and NPR1 cooperatively inhibit invasion of Pst DC3000/avrRps4

The mRNA expression levels of *ATG6* in Col were significantly increased after 6 h, 12 h and 24 h under *Pst* DC3000/*avrRps4* treatment (**Fig. 7a**). Similarly, both the *ATG6* gene and protein were significantly up-regulated under 0.5 mM SA treatment (**Fig. 7b and c**). These results suggest that the expression of *ATG6* could be induced by *Pst* DC3000/*avrRps4* and 0.5 mM SA treatment.

Considering that ATG6 increases NPR1 protein levels (**Fig. 5a-b**) and promotes its nuclear accumulation (**Fig. 3**), as well as maintains NPR1 stability (**Fig. 6**), then we studied the role of ATG6-NPR1 interactions in plant immunity. However, studying the function of ATG6 is challenging due to the lethality of homozygous *atg6* mutant (Qin *et al*., 2007; Harrison-Lowe and Olsen, 2008; Patel and Dinesh-Kumar, 2008). According to our previous report (Lei *et al*., 2020; Zhang *et al*., 2023), *ATG6* was silenced using artificial miRNA^{ATG6} (amiRNA^{ATG6}) delivered by the gold nanoparticles (AuNPs). First, the effect of *ATG6* silencing in Col on the plant immune response was investigated. Similar to *atg5*, Col/silencing *ATG6* exhibited more active growth of *Pst* DC3000/*avrRps4* than Col/negative control (NC) after *Pst* DC3000/*avrRps4* infiltration for 3 days (**Fig. 7d**). Furthermore, according to the previously reported methods (Ohira *et al*., 2017; Gomez *et al*., 2022), we generated two amiRNA^{ATG6}lines (amiRNA^{ATG6} # 1 and amiRNA^{ATG6} # 2) designed against *ATG6* and placed under the control of a β-estradiol inducible promoter. There were no significant phenotypic differences in amiRNA^{ATG6} # 1 compared to the Col, while amiRNA^{ATG6} # 2 exhibited a slight leaf developmental defect (**Fig. 7e** and **f**). Subsequently, we investigated the expression of *ATG6* following treatment with 100 μM β-estradiol. Our results showed that, after 100 μM β-estradiol treatment for 1~3 d, the expression of *ATG6* in both amiRNA^{ATG6} # 1 and amiRNA^{ATG6} # 2 lines was significantly lower than that in Col. Specifically, the expression of *ATG6* in the amiRNA^{ATG6} *#1* and amiRNA^{ATG6} *#2* lines decreased by 50~70% compared with Col (**Fig. 7g**). Furthermore, to assess the function of ATG6 in plant immune, we performed infiltrations of *Pst* DC3000/*avrRps4* after 100 uM β-estradiol treatment for 24 h. We compared the growth of *Pst* DC3000/*avrRps4* in the amiRNA^{ATG6} lines and Col. The results clearly demonstrate that the growth of *Pst* DC3000/*avrRps4* in amiRNA^{ATG6} # 1 and amiRNA^{ATG6} # 2 was significantly more compared to Col (**Fig. 7h**). Moreover, we silenced *ATG6* in *NPR1-GFP* (*NPR1-GFP*/silencing *ATG6*), and *NPR1-GFP/atg5* (crossed *NPR1-GFP* with *atg5* to obtain *NPR1-GFP/atg5*) was used as an autophagy-deficient control. There was more *Pst* DC3000/*avrRps4* growth in *NPR1-GFP*/silencing *ATG6* and *NPR1-GFP/atg5* compared to *NPR1-GFP*/NC after *Pst* DC3000/*avrRps4* infiltration (**Fig. 7i**). In contrast, the growth of *Pst* DC3000/*avrRps4* in *NPR1- GFP*, *ATG6-mCherry*, *ATG6-mCherry* × *NPR1-GFP* was significantly lower than that in Col and *npr1* (**Fig. 7j**) and was the lowest in *ATG6-mCherry* × *NPR1-GFP* (**Fig. 7j**).

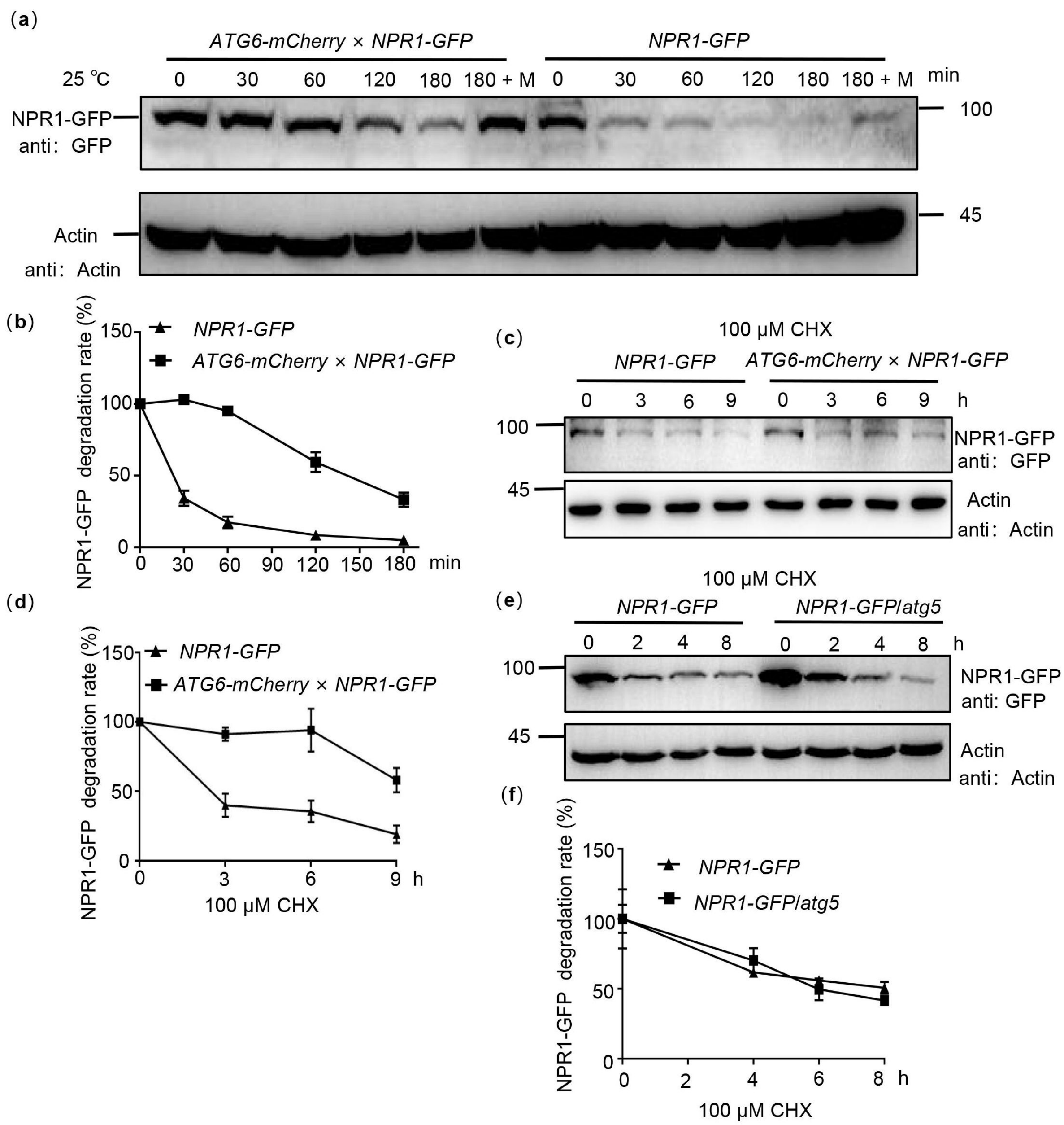

Figure 6.

ATG6 improves the protein stability of NPR1.

(a). NPR1-GFP degradation assay in *Arabidopsis.* Total proteins from 7-day-old seedlings of *NPR1-GFP* and *ATG6-mCherry* x *NPR1-GFP* were extracted, using Actin as an internal reference. “M” indicates 100 μM MG 115 treatment. **(b).** Quantification of NPR1-GFP degradation rates in **(a)** using Image J. In **(a and b),** the extracts were incubated for 0 ~ 180 min at room temperature (25°C), the degradation rate of NPR1-GFP was analyzed. **(c).** NPR1-GFP protein turnover. 7-day-old *NPR1-GFP* and *ATG6-mCherry* x *NPR1-GFP* seedlings were treated with 100 μM cycloheximide (CHX) for different times. Total proteins were analyzed, Actin was used as an internal reference. **(d).** Quantification of NPR1-GFP protein turnover rates in **(c)** using Image J. **(e).** NPR1-GFP protein turnover.7-day-old *NPR1-GFP* and *NPR1-GFP/atg5* seedlings were treated with 100 μM cycloheximide (CHX) for different times. Total proteins were analyzed, Actin was used as an internal reference. **(f).** Quantification of protein levels of NPR1-GFP in **(e)** using Image J. All experiments were performed with three biological replicates.

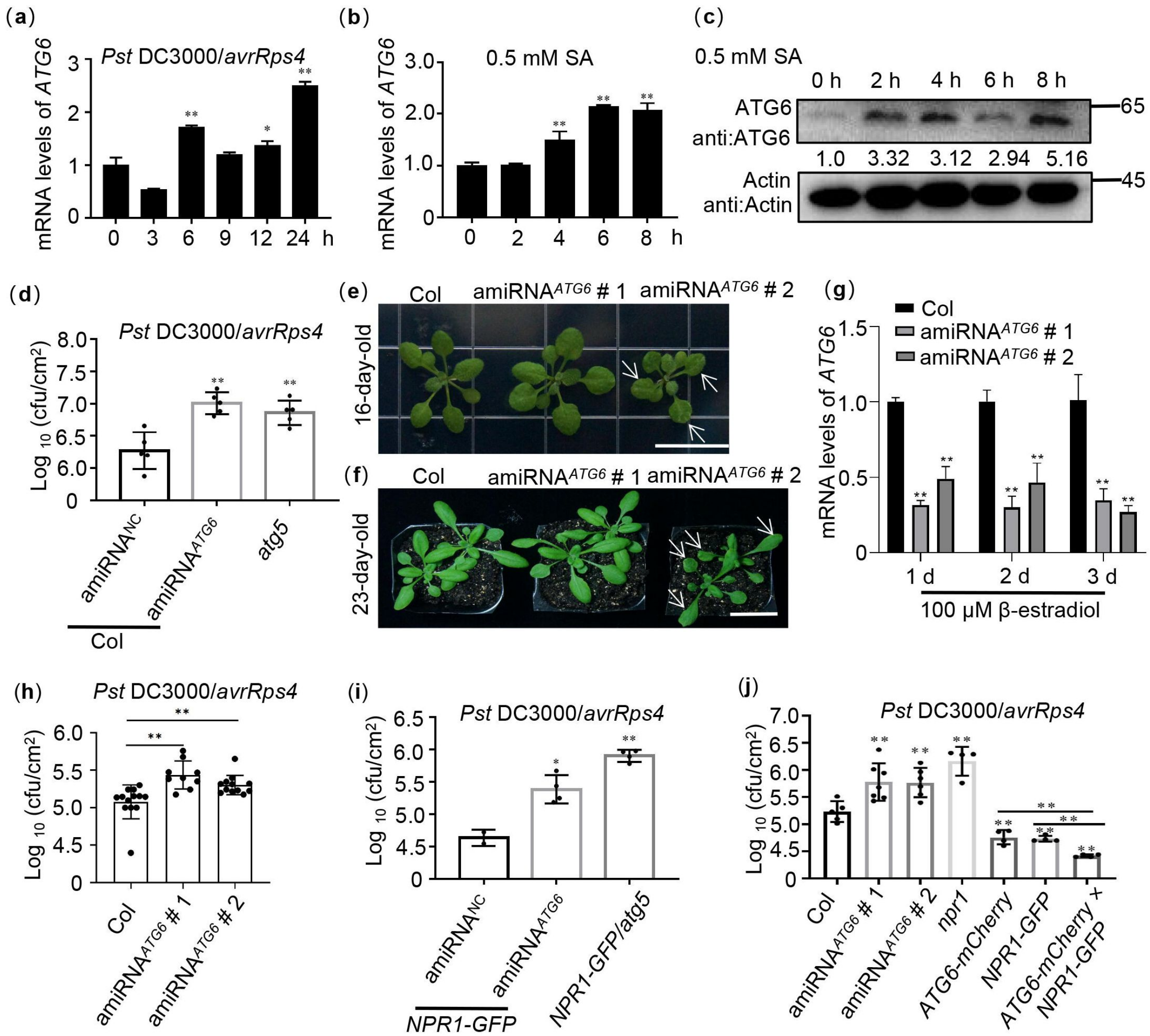

Figure 7.

ATG6 and NPR1 cooperatively inhibit the growth of *Pst DC3000/avrRps4.*

(a). Expression of *ATG6* under *Pst DC3000/avrRps4* injection in 3-week-old Col leaves. **(b).** Expression of *ATG6* in the presence of 0.5 mM SA in 3-week-old Col leaves. **(c).** The protein levels of ATG6 after 0.5 mM SA in 3- week-old Col leaves. Total leaf proteins from *Arabidopsis* were analyzed, Actin was used as an internal reference. Numerical values indicate quantitative analysis of ATG6 protein using image J. **(d).** Growth of *Pst DC3000/avrRps4* in Col/silencing *ATG6* and Col/negative control (NC). **(e).** Phenotypes of 16-day old amiRNAATG^{6}# 1 and amiRNAATG^{6}# 2. Bar, 1 cm. **(f).** Phenotypes of 23-day-old amiRNAATG^{6}# 1 and amiRNAATG^{6}# 2. Bar, 3 cm. **(g).** Expression of *ATG6* in Col, amiRNAATG^{6}# 1 and amiRNAATG^{6}# 2 under injection treatment of 100 µM -estradiol. **(h).** Growth of *Pst DC3000/avrRps4* in *Arabidopsis* leaves of amiRNAATG^{6}# 1, amiRNAATG^{6}# 2 and Col. (i).Growth of *Pst DC3000/avrRps4* in NPR1-GFP/silencing *ATG6* and *NPR1-GFPINC.* **(j).** Growth of *Pst DC3000/avrRps4* in *Arabidopsis* leaves of Col, amiRNAATG^{6}# 1, amiRNAATG^{6}# 2, *npr1, NPR1-GFP, ATG6-mCherry* and *ATG6-mCherry* x *NPR1-GFP.* In (d, h-j), a low dose of *Pst DC3000/avrRps4* (OD_{600} = 0.001) was infiltrated. After 3 days, the growth of *Pst DC3000/avrRps4* was counted. *or** indicates that the significant difference compared to the control is at the level of 0.05 or 0.01 (Student t test p value,* p< 0.05 or** p< 0.01). All experiments were performed with three biological replicates.

These results confirm that ATG6 and NPR1 cooperatively enhance *Arabidopsis* resistance to inhibit *Pst* DC3000/*avrRps4* invasion. Together, these results suggest that ATG6 improves plant resistance to pathogens by regulating NPR1.

Discussion

Although SA signaling and autophagy are related to the plant immune system (Yoshimoto *et al.*, 2009; Munch *et al.*, 2014; Wang *et al.*, 2016), the connection of these two processes in plant immune processes and their interaction is rarely reported. Previous studies have shown that unrestricted pathogen-induced PCD requires SA signalling in autophagy-deficient mutants. SA and its analogue benzo (1,2,3) thiadiazole-7-carbothioic acid (BTH) induce autophagosome production (Yoshimoto *et al.*, 2009). Moreover, autophagy has been shown to negatively regulates *Pst* DC3000/*avrRpm1*-induced PCD via the SA receptor NPR1 (Yoshimoto *et al.*, 2009), implying that autophagy regulates SA signaling through a negative feedback loop to limit immune- related PCD. Here, we demonstrated that ATG6 increases NPR1 protein levels and nuclear accumulation (**Fig. 3** and **5**). Additionally, ATG6 also maintains the stability of NPR1 and promotes the formation of SINCs-like condensates (**Fig. 5** and **6**). These findings introduce a novel perspective on the positive regulation of NPR1 by ATG6, highlighting their synergistic role in enhancing plant resistance.

Our results confirmed that *ATG6* overexpression significantly increased nuclear accumulation of NPR1 (**Fig. 3**). ATG6 also increases NPR1 protein levels and improves NPR1 stability (**Fig. 5 and 6**). Therefore, we consider that the increased nuclear accumulation of NPR1 in *ATG6-mCherry* x *NPR1-GFP* plants might result from higher levels and more stable NPR1 rather than the enhanced nuclear translocation of NPR1 facilitated by ATG6. To verify this possibility, we determined the ratio of NPR1-GFP in the nuclear localization versus total NPR1-GFP. Notably, the ratio (nucleus NPR1/total NPR1) in *ATG6-mCherry* × *NPR1-GFP* was not significantly different from that in *NPR1- GFP*, and there is a similar phenomenon in *N. benthamiana* (**Fig. 3c-f**). Further we analyzed whether ATG6 affects NPR1 protein levels and protein stability. Our results show that ATG6 increases NPR1 protein levels under SA treatment and ATG6 maintains the protein stability of NPR1 (**Fig. 5 and 6**). These results suggested that the increased nuclear accumulation of NPR1 by ATG6 result from higher levels and more stable NPR1.

NPR1 is an important signaling hub of the plant immune response. Nuclear localization of NPR1 is essential to enhance plant resistance (Kinkema *et al.*, 2000; Chen *et al.*, 2021b), it interacts with transcription factors such as TGAs in the nucleus to activate expression of downstream target genes (Chen *et al.*, 2019; Chen *et al.*, 2021a). A recent study showed that nuclear-located ATG8h recognizes C1, a geminivirus nuclear protein, and promotes C1 degradation through autophagy to limit viral infiltration in solanaceous plants (Li *et al.*, 2020). Here, we confirmed that ATG6 is also distributed in the nucleus and ATG6 is co-localized with NPR1 (**Fig. 1d** and **2**), suggesting that ATG6 interact with NPR1 in the nucleus. ATG6 synergistically inhibits the invasion of *Pst* DC3000/*avrRps4* with NPR1. Chen et al. found that in the nucleus, NPR1 can recruit enhanced disease susceptibility 1 (EDS1), a transcriptional coactivator, to synergistically activate expression of downstream target genes (Chen *et al.*, 2021a). Previous studies have shown that acidic activation domains (AADs) in transcriptional activators (such as Gal4, Gcn4 and VP16) play important roles in activating downstream target genes. Acidic amino acids and hydrophobic residues are the key structural elements of AAD (Pennica *et al.*, 1984; Cress and Triezenberg, 1991; Van Hoy *et al.*, 1993). Chen et al. found that EDS1 contains two ADD domains and confirmed that EDS1 is a transcriptional activator with AAD (Chen *et al.*, 2021a). Here, we also have similar results that *ATG6* overexpression significantly enhanced the expression of *PR1* and *PR5* (**Fig. 4b-c and Fig.S10**), and that the ADD domain containing acidic and hydrophobic amino acids is also found in ATG6 (148-295 AA) (**Fig. S15**). We speculate that ATG6 might act as a transcriptional coactivator to activate *PRs* expression synergistically with NPR1.

A recent study showed that SA not only enhances plant resistance by increasing NPR1 nuclear import and transcriptional activity, but also promotes cell survival by coordinating the distribution of NPR1 in the nucleus and cytoplasm (Zavaliev *et al*., 2020). Notably, NPR1 accumulated in the cytoplasm recruits other immunomodulators (such as EDS1, PAD4 etc) to form SINCs to promote cell survival (Zavaliev *et al*., 2020). Similarly, we also found that NPR1 accumulated abundantly in the cytoplasm after SA treatment and that ATG6 significantly increased NPR1 protein levels (**Fig. 3c, 3e and 5a-b**). Obviously, the accumulation of NPR1 in the cytoplasm may be related to ATG6 synergizing with NPR1 to enhance plant resistance. Interestingly, *ATG6* overexpression significantly increased the formation of SINCs-like condensates (**Fig. 5c-d** and **Supplemental movie 1-2**), which should also be a way for ATG6 and NPR1 to synergistically resist invasion of pathogens. We consider that ATG6 promotes the formation of SINCs-like condensates through the dual action of endogenous and exogenous SA. Considering that ATG6 promotes SINCs-like condensates formation, we further examined changes in cell death in Col, amiRNA^{ATG6}# 1, amiRNA^{ATG6} # 2, *npr1*, *NPR1- GFP*, *ATG6-mCherry* and *ATG6-mCherry* × *NPR1-GFP* plants. The results of Taipan blue staining showed that *Pst* DC3000/*avrRps4*-induced cell death in *npr1*, amiRNA^{ATG6} # 1 and amiRNA^{ATG6} # 2 was significantly higher compared to Col (**Fig. S16**). Conversely, *Pst* DC3000/*avrRps4*-induced cell death in *ATG6-mCherry*, *NPR1-GFP* and *ATG6-mCherry* × *NPR1-GFP* was significantly lower compared to Col. Notably, *Pst* DC3000/*avrRps4*-induced cell death in *ATG6-mCherry* × *NPR1-GFP* was significantly lower compared *ATG6-mCherry* and *NPR1-GFP* (**Fig. S16**). These results suggest that ATG6 and NPR1 cooperatively inhibit *Pst* DC3000/*avrRps4*-induced cell dead.

ATG6 is a common and required subunit of PtdIns3K lipid kinase complexes, which regulates autophagosome nucleation in *Arabidopsis* (Qi *et al*., 2017; Bozhkov, 2018). In this study, we also found that ATG6 can maintain the stability of NPR1. Thus, to confirm whether the regulation of NPR1 protein stability by ATG6 is autophagy-dependent, we used autophagy inhibitors (Concanamycin A, ConA and Wortmannin, WM) to detect the degradation of NPR1-GFP. Cell-free degradation assays showed that 100 μM MG115 treatment significantly inhibited the degradation of NPR1-GFP. However, 5 μM concanamycin A treatment did not significantly delay NPR1 degradation (**Fig. S17**). Remarkably, treatment with 30 μM Wortmannin resulted in a slight acceleration of NPR1 degradation, while the combined treatment of ConA and WM significantly expedited the degradation of NPR1 (**Fig. S17**). This may be related to crosstalk between autophagy and 26S Proteasome. It has been demonstrated that autophagy directly regulates the activity of the 26S proteasome under normal conditions or treatment with *Pst* DC3000 (Marshall *et al*., 2015; Ustun *et al*., 2018). Marshall et al. found that the 26S proteasome subunits (RPN1, RPN3, RPN5, RPN10, PAG1, PBF1) are significantly enriched in autophagy-deficient mutantsunder normal growth conditions (Marshall *et al*., 2015). Treatment with concanamycin A (ConA), an inhibitor of vacuolar-type ATPase, increased the level of the 20S proteasome subunit PBA1 under treatment with *Pst* DC3000 (Ustun *et al*., 2018). In addition, we also analyzed the degradation of NPR1-GFP in *NPR1-GFP* and *NPR1-GFP*/*atg5* following 100 μM cycloheximide (CHX) treatment. The results show that the degradation rate of NPR1-GFP in *NPR1-GFP*/*atg5* plants was similarly to that in *NPR1-GFP* plants (**Fig. 6e and f**). These results suggest that deletion of ATG5 do not affect the protein stability of NPR1.

An increasing number of studies have shown that ATGs differentially affect plant immunity. Deletion of ATGs (ATG5, ATG7, ATG10 etc.) leads to reduced resistance of plants to necrotrophic pathogens (Lai *et al*., 2011; Lenz *et al*., 2011; Minina *et al*., 2018). ATGs can directly interact with other proteins to positively regulate plant immunity. In *N. benthamiana*, ATG8f interacts the effector protein βC1 of the cotton *leaf curl multan virus* and promotes its degradation to limit pathogen invasion (Haxim *et al*., 2017). Notably, ATG18a can interact with WRKY33 transcription factor to synergistically against *Botrytis* invasion (Lai *et al*., 2011). Our evidence shows that ATG6 interacts with NPR1 and works together to counteract pathogen invasion by positively regulating NPR1 and SA levels *in vivo*. In conclusion, we unveil a novel relationship in which ATG6 positively regulates NPR1 in plant immunity (**Fig. 8**). ATG6 interacts with NPR1 to synergistically enhance plant resistance by regulating NPR1 protein levels, stability, nuclear accumulation, and formation of SINCs-like condensates.

Material and Methods

Plasmid construction

Details of plasmid construction methods are listed in Methods S1 of Supplemental Data 1, primer used are listed in Table S1 and S2 of Supplemental Data 1. The mapping of vectors is listed in Supplemental Data 2.

Plant material Arabidopsis

35S::NPR1-GFP (in *npr1-2* background) and *npr1-1* were kindly provided by Dr. Xinnian Dong of Duke University; *atg5-1* (SALK_020601).

UBQ10::ATG6-mCherry, *UBQ10::ATG6-GFP and* amiRNA^{ATG6} lines were obtained by *Agrobacterium* transformation (Clough and Bent, 1998). *ATG6*, *NPR1* double overexpression of *Arabidopsis* (*ATG6-mCherry* × *NPR1-GFP*) and *NPR1-GFP*/*atg5* were obtained by crossing respectively.

Full description of the *Arabidopsis* screening is included Methods S2 of Supplemental Data 1. Details of plant material are listed in Table S3 of Supplemental Data 1.

Growth conditions Arabidopsis thaliana

All *Arabidopsis thaliana* (*Arabidopsis*) seeds were treated in 10% sodium hypochlorite for 7 min, washed with ddH_2O, and treated in 75% ethanol for 30 s, finally washed three times with ddH_2O. Seeds were sown in 1/2 MS with 2% sucrose solid medium, vernalized at 4℃ for 2 days.

For 7-day-old *Arabidopsis* seedling cultures, the plates were placed under the following conditions: daily cycle of 16 h light (~80 μmolm^{-2}. s^{-1}) and 8 h dark at 23 ± 2℃.

For 3-week-old *Arabidopsis* cultures, after 7 days of growth on the plates, the seedlings were transferred to soil for further growth for 2 weeks under the same conditions (Zhang *et al*., 2018).

N. benthamiana

For 3-week-old *N. benthamiana* cultures, seeds were sown in the soil and vernalized at 4℃ for 2 days. After 10 days of growth on soil, the seedlings were transferred to soil for further growth for 2 weeks under the same conditions (Jiao *et al*., 2019).

Treatment conditions

Treatment of 7-day-old seedlings

For SA treatment

7-day-old *Arabidopsis* seedlings were transferred to 1/2 MS liquid medium containing 0.5 mM SA for 0, 3 and 6 h, respectively. The corresponding results are shown in **Fig. 2f**, g, **3c**, d and 5a.

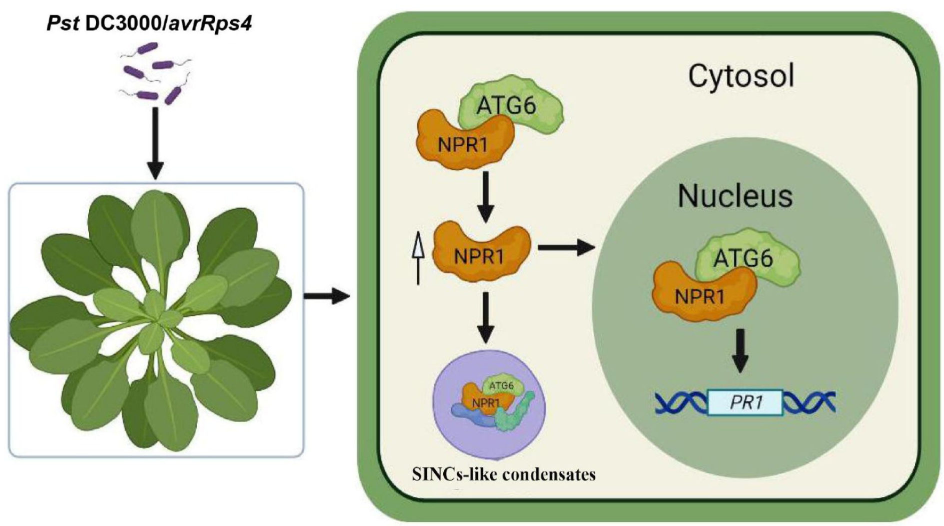

Figure 8.

Working model for NPR1 regulation by ATG6.

ATG6 interacts directly with NPR1 to increase NPR1 protein level and stability, thereby promoting the formation of SINCs-like condensates and increasing the nuclear accumulation of NPR1. ATG6 synergistically activates *PRs* expression with NPR1 to cooperatively enhance resistance to inhibit *Pst DC3000/avrRps4* invasion in *Arabidopsis*.

For cycloheximide (CHX) treatment

Seedlings of *Arabidopsis* (7 days) were transferred to 1/2 MS liquid medium containing 100 μM CHX for 0, 3, 6 and 9 h, respectively. The corresponding results are shown in **Fig. 6c and e**.

Treatment of 3-week-old Arabidopsis

For silencing ATG6 in Col and NPR1-GFP

As previously described (Lei *et al*., 2020; Zhang *et al*., 2022; Zhang *et al*., 2023), 1 mM gold nanoparticles (AuNPs) were synthesized. The artificial microRNA (amiRNA)^{ATG6} (UCAAUUCUAGGAUAACUGCCC) was designed based on the Web MicroRNA Designer (*http://wmd3.weigelworld.org/*) platform. The complementary sequence of amiRNA^{ATG6} is located on the eighth exon of the *ATG6* gene. The sequence of "UUCUCCGAACGUGUCACGUTT" was used as a negative control (NC). NC is a universal negative control without species specificity (Gao *et al*., 2018; Lei *et al*., 2020). amiRNA^{ATG6} and amiRNA^{NC} synthesized by Suzhou GenePharma. AuNPs (1 mM) and amiRNA^{ATG6} (20 μM) were incubated at a 9:1 ratio for 30 min at 25°C, 50 rpm. After incubation, a mixture of AuNPs and amiRNA^{ATG6}was diluted 15-fold with the infiltration buffer (pH 5.7, 10 mM MES, 10 mM $MgCl_2$) and infiltrated through the abaxial leaf surface into 3-week-old Col or *NPR1-GFP* for 1-3 days. The third day was chosen as material for *ATG6* silencing. After the third day of AuNPs-amiRNA^{ATG6}and AuNPs-amiRNA^{NC} infiltration, *Pst* DC3000/*avrRps4* was infiltrated, and then growth of *Pst* DC3000/*avrRps4* was detected.

For β-estradiol treatment

100 μM β-estradiol was infiltrated to treat 3-week-old *Arabidopsis* leaves. After 24 h of treatment with β-estradiol, *Pst* DC3000/*avrRps4* was infiltrated and then growth of *Pst* DC3000/*avrRps4* was detected after 3 d.

For Pst DC3000/avrRps4 infiltration

Infiltration with *Pst* DC3000/*avrRps4* was performed as previously described (Wang *et al*., 2016; Skelly *et al*., 2019). Full description of the *Pst* DC3000/*avrRps4* culture is included in Methods S3 of Supplemental Data 1.

For SA treatment

For 3-week-old Col, 0.5 mM SA was infiltrated into the leaves for 0, 2, 4, 6 and 8 h. The corresponding results are shown in **Fig. 7b and c**.

Yeast two-hybrid assay

Yeast two-hybrid experiments were performed according to the previously described protocol (Fu *et al*., 2012). Full description of Yeast two-hybrid is included in Methods S4 of Supplemental Data 1.

Pull down assays in vitro

500 μL of GST, GST-ATG6, SnRK2.8-GST were incubated with GST-tag Purification Resin (Beyotime, P2250) for 2 h at 4°C. The mixture was then centrifuged at 1500 g for 1 min at 4°C, and the resin was washed three times with PBS buffer. Next, the GST-tag purification resin was incubated with the NPR1-His for 2 h at 4°C. After washing three times with PBS buffer, 2 × sample buffers were added to the resin and denatured at 100 °C for 10 mins. The resulting samples were then used for western blotting analysis. Full description of prokaryotic proteins expression is included in Methods S5 of Supplemental Data 1.

Co-immunoprecipitation

0.5 g leaves of *N. benthamiana* transiently transformed with ATG6-mCherry + GFP and ATG6-mCherry + NPR1-GFP were fully ground in liquid nitrogen and homogenized in 500 μL of lysis buffer (50 mM Tris-HCl pH 7.5, 150 mM NaCl, 0.5 mM EDTA, 5% Glycerol, 0.2% NP40, 1 mM PMSF, 40 μM MG115, protease inhibitor cocktail 500× and phosphatase inhibitor cocktail 5000×). The samples were then incubated on ice for 30 mins, and centrifuged at 10142 g (TGL16, cence, hunan, China) for 15 mins at 4°C. The supernatant (500 μL) was incubated with 20 μL of GFP-Trap Magnetic Agarose beads (ChromoTek, gtma-20) in a 1.5 mL Eppendorf tube for 2 h by rotating at 4°C. After incubation, the GFP-Trap magnetic Agarose beads were washed three times with cold wash buffer (50 mM Tris-HCl pH 7.5, 150 mM NaCl, 0.5 mM EDTA) and denatured at 75°C for 10 minutes after adding 2 × sample buffer. Western blotting was performed with antibodies to ATG6 and GFP.

Nuclear and cytoplasmic separation

Nuclear and cytoplasmic separation were performed according to the previously described method (Kinkema *et al*., 2000). Full description of nuclear and cytoplasmic separation is given in Methods S6 of Supplemental Data 1.

Protein degradation in vitro

Protein degradation assays were performed according to a previously described method (Spoel *et al*., 2009; Saleh *et al*., 2015). Full description of protein degradation is included in Methods S7 of Supplemental Data 1.

Protein Extraction and Western Blotting Analysis

Protein extraction and western blotting were performed as previously described (Lei *et al*., 2020; Zhang *et al*., 2022). Protein was denatured at 100°C for 10 mins. NPR1 protein was denatured at 75°C for 10 mins (Lei *et al*., 2020). Full description is included in Methods S8 of Supplemental Data 1. Antibody information is presented in Table S4 of Supplemental Data 1.

Confocal microscope observation

For nuclear localization of NPR1-GFP observation

7-day-old seedlings of *NPR1-GFP* and *ATG6-mCherry* × *NPR1-GFP* were sprayed with 0.5 mM SA for 0 and 3 h. GFP and mCherry fluorescence signals in leaves were observed under the confocal microscope (Zeiss LSM880). Statistical data were obtained from three independent experiments, each comprising five individual images, resulting in a total of 15 images analyzed for this comparison.

For the Bimolecular Bluorescence Complementation assay (BiFC)

Agrobacterium was infiltrated into *N. benthamiana* as previously described (Jiao *et al*., 2019). Fluorescence signals were observed after 3 days. The full description of BiFC is contained in Methods S9 and S10 of Supplemental Data 1.

For the observation of SINCs-like condensates

Agrobacterium was infiltrated into *N. benthamiana*. After 2 days, the leaves were treated in 1 mM SA solution for 24 h, and then fluorescence signals were observed. At least 20 image sets were obtained and analyzed. A full description of SINCs-like condensates observation is included in Methods S11 of Supplemental Data 1.

For growth of Pst DC3000/avrRps4

A low dose (OD_{600} = 0.001) of *Pst* DC3000/*avrRps4* was used for the infiltration experiments. After 3 days, the colony count was counted according to a previous description (Wang *et al*., 2016; Lei *et al*., 2020). Full description of the growth of *Pst* DC3000/*avrRps4* is given in Methods S12 of Supplemental Data 1.

Free SA measurement

Free SA was extracted from 3-week-old *Arabidopsis* using a previously described method (Wang *et al*., 2016; Gong *et al*., 2020). Free SA was measured by High-performance liquid chromatography (Shimadzu LC-6A, Japan). Detection conditions: 294 nm excitation wavelength, 426 nm emission wavelength.

Real-Time Quantitative PCR (RT-qPCR)

Total RNA was extracted from *Arabidopsis* (100 mg) using Trizol RNA reagent (Invitrogen, 10296-028, Waltham, MA, USA). RT-qPCR assays were performed as previously described (Zhang *et al*., 2018; Zhang *et al*., 2022). All primers for RT-qPCR are listed individually in Table S5 of Supplemental Data 1. Full description of RT-qPCR is included in Methods S13 of Supplemental Data 1.

Trypan Blue Staining

The leaves of 3-week-old Col, amiRNA^{ATG6} # 1, amiRNA^{ATG6} # 2, *npr1*, *NPR1- GFP*, *ATG6-mCherry* and *ATG6-mCherry* × *NPR1-GFP* plants, located in the fifth and sixth positions, were infiltrated with *Pst* DC3000/*avrRps4*. After 3 days, the leaves were excised and subjected to a 1 min boiling step in trypan blue staining buffer (consisting of 10 g phenol, 10 mL glycerol, 10 mL lactic acid, 10 mL ddH_2O, and 10 mg trypan blue), followed by destaining 3 times at 37°C in 2.5 mg/mL chloral hydrate.

Statistical Analysis

All quantitative data in this study were presented as mean ± SD. The experimental data were analyzed by a two-tailed Student's *t*-test. Significance was assigned at *P* values < 0.05 or < 0.01.

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Author contribution

B.-H.Z., J.Z. and W.-L.C. designed and supervised the research; B.-H.Z. and S.-Q.H. performed majority experiments; B.-H.Z., Y.-X.M. and H.C. synthesized gold nanoparticles (AuNPs); B.-H.Z. and S.-Q.H., Y.-Z.T. performed western blotting analysis; B.-H.Z.,Y.Z and X.L performed confocal experiment; B.-H.Z. and S.-Q.H. analyzed the data; B.-H.Z. and S.-Q.H. prepared figures; B.-H.Z. and S.-Q.H. wrote the manuscript; J.Z. and W.-L.C. supervised and completed the writing; S.-Y.G. drew working model and edited language. J.Z. and W.-L.C. provided reagents/materials/analysis tools; J.Z. and W.-L.C. agrees to serve as the corresponding author responsible for contact and ensures communication.

Conflicts of interest

The author declares that there are no conflicts of interest.

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Data availability

The authors confirm that the data supporting the results of this study are available in the article and its supplementary materials.

Supplementary figures

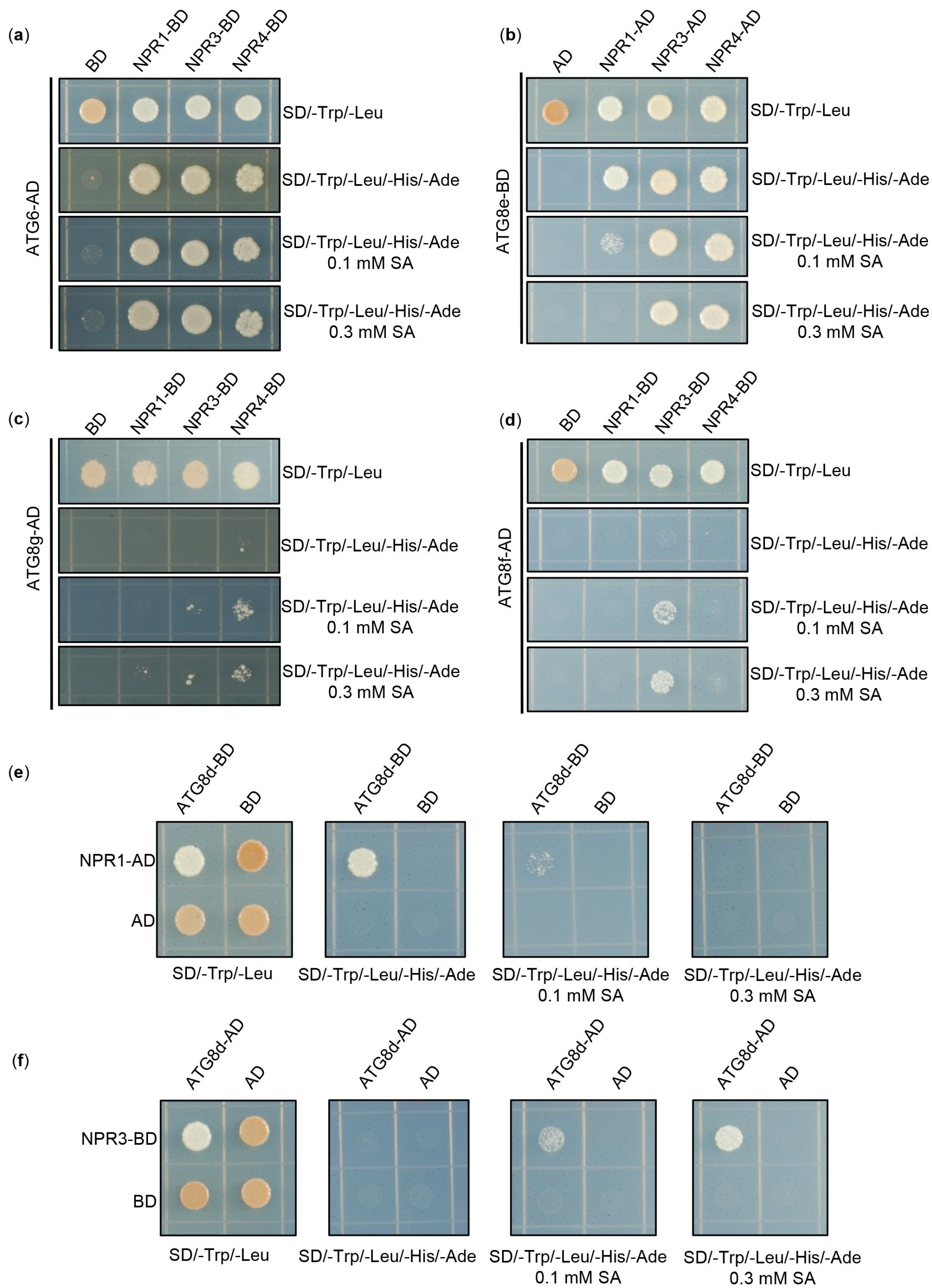


Fig. S1 Physical interaction between NPRs and ATGs in yeast. The CDS of *NPR1*, *NPR3*, *NPR4*, *ATG6* and *ATG8d-g* were fused to the AD and BD domains, respectively, and co-expressed with yeast cells. Yeast cell growth on SD/-Trp-Leu-His-Ade media represents interaction. Add different concentrations of SA to SD/-Trp-Leu-His-Ade medium to verify the effect of SA on their interactions. All experiments were performed with three biological replicates.

Supplemental Figure 1

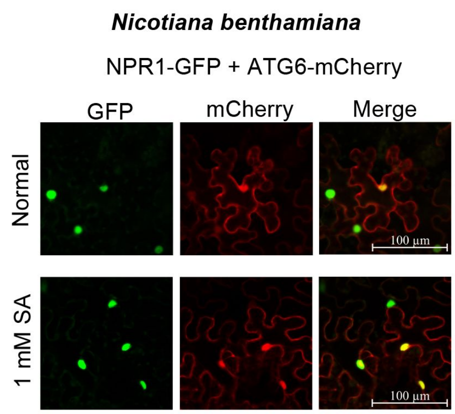


Fig. S2 Co-localization of NPR1-GFP and ATG6-mCherry in *N.benthamiana* under normal and SA treatment conditions. NPR1-GFP were co-expressed with ATG6-mCherry in *N. benthamiana* for 2 d followed by confocal observation. Scale bar, 100 μm. Experiment was performed with three biological replicates.

Supplemental Figure 2

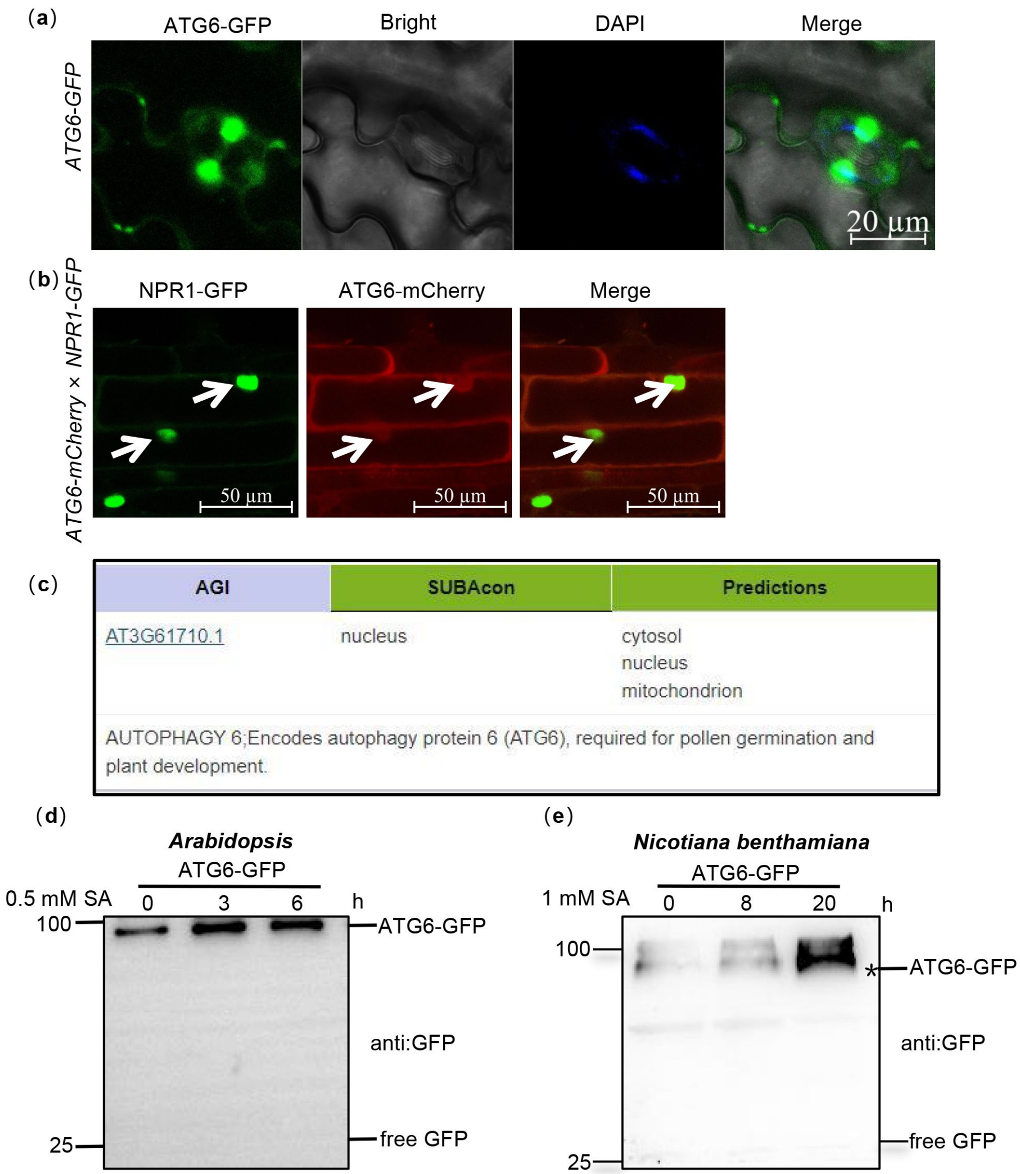


Fig. S3 The nuclear localization of ATG6 in *Arabidopsis*. (a). The nuclear localization of ATG6-GFP in *ATG6-GFP* under normal condition. Scale bar, 20 μm. **(b).** The nuclear localization of ATG6-mCherry in *ATG6-mCherry* × *NPR1-GFP*. Scale bar,50 μm. **(c).** Predicted subcellular localization of ATG6 by *Arabidopsis* database (https://suba.live/). **(d).** ATG6-GFP and free GFP protein levels in *ATG6-GFP Arabidopsis*. **(e).** ATG6-GFP and free GFP protein levels in *N. benthamiana*. Black asterisk (*) indicate ATG6-GFP bands. All experiments were performed with three biological replicates.

Supplemental Figure 3

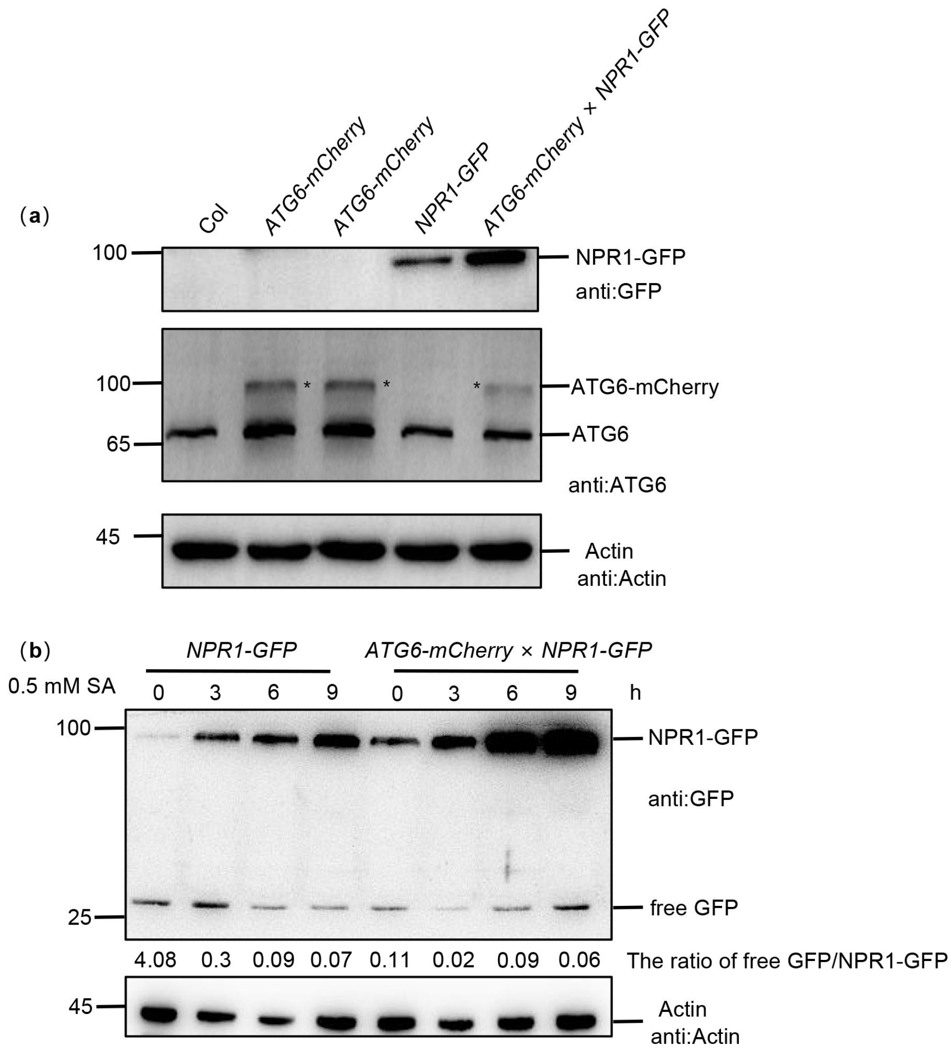


Fig. S4 Identification of *ATG6-mCherry* × *NPR1-GFP* plants. (a). Total proteins from 7-day-old seedlings were extracted. Western blots analysis with ATG6 and GFP antibodies. Actin was used as an internal reference. Black asterisk (*) indicate ATG6-mCherry bands. **(b).** NPR1-GFP and free GFP protein levels in 7-day-old seedlings of *NPR1-GFP* and *ATG6-mCherry* × *NPR1-GFP* plants after 0.5 mM SA treatment for 0, 3, 6 and 9 h. Numerical values indicate the ration of NPR1-GFP/free GFP. All experiments were performed with three biological replicates.

Supplemental Figure 4

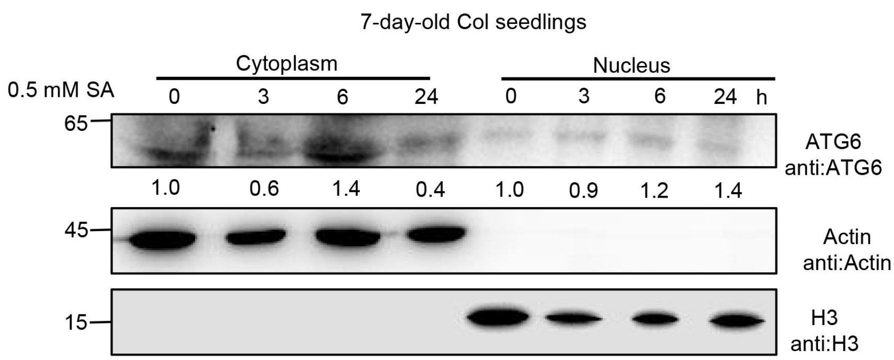


Fig. S5 Subcellular fractionation of endogenous ATG6 in Col after 0.5 mM SA treatment for 0, 3, 6 and 20 h. Cytoplasmic and nuclear proteins were extracted from *Arabidopsis*. Endogenous ATG6 were detected using ATG6 antibody. Actin and H3 were used as cytoplasmic and nucleus internal reference, respectively. Numerical values indicate quantitative analysis of ATG6 using image J. Experiment was performed with three biological replicates.

Supplemental Figure 5

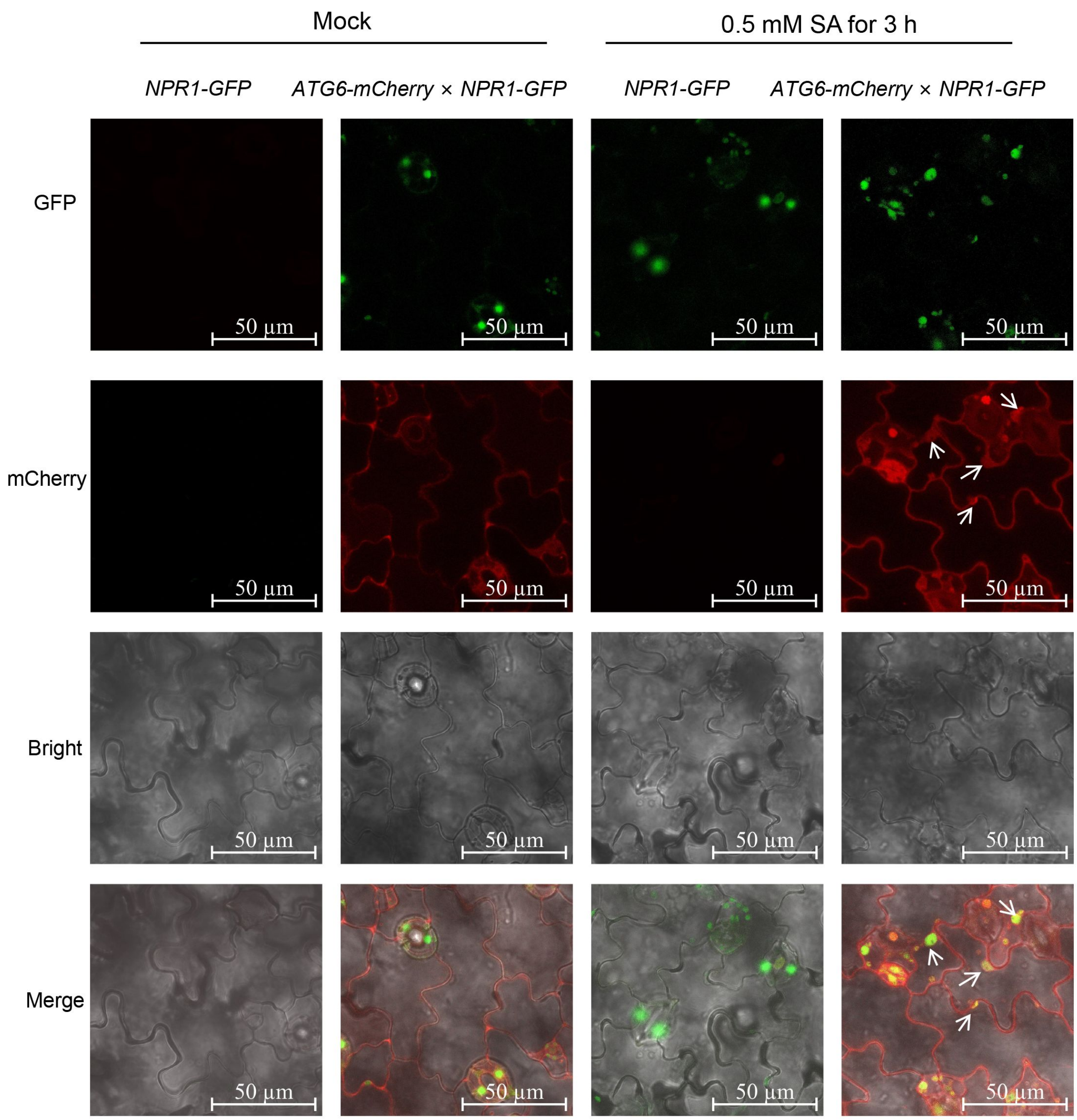


Fig. S6 Confocal images of NPR1-GFP nuclear localization in 7-day-old seedlings of *NPR1-GFP* and *ATG6-mCherry* × *NPR1-GFP* under normal and 0.5 mM SA spray for 3 h. Scale bar, 50 μm.

Supplemental Figure 6

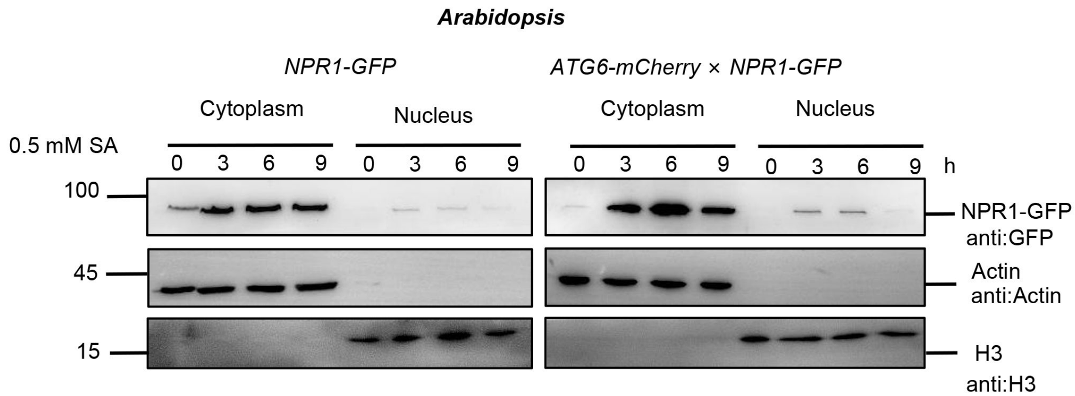


Fig 3c repeat 2

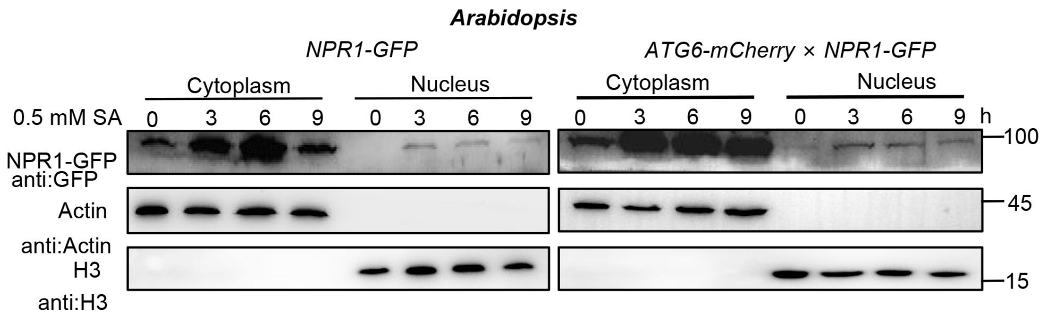


Fig 3c repeat 3

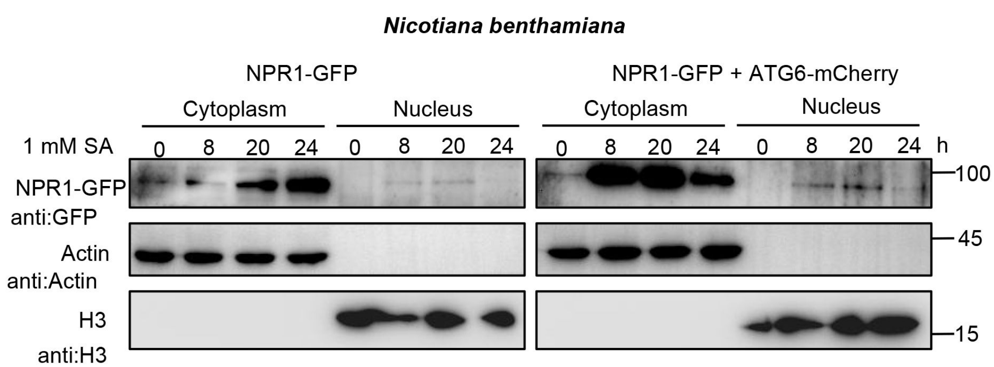


Fig 3e repeat 2

Fig. S7. ATG6 increases the nuclear accumulation of NPR1 under SA treatment. Replicate experiments at Fig 3c and 3e. (Fig 3c). Subcellular fractionation of NPR1-GFP in 7-day-old seedlings of *NPR1-GFP* and *ATG6-mCherry* × *NPR1-GFP* after 0.5 mM SA treatment for 0, 3 and 6 h. **(Fig 3e).** Subcellular fractionation of NPR1-GFP in *N. benthamiana* after 1 mM SA treatment for 0, 8,20 and 24 h.

Supplemental Figure 7

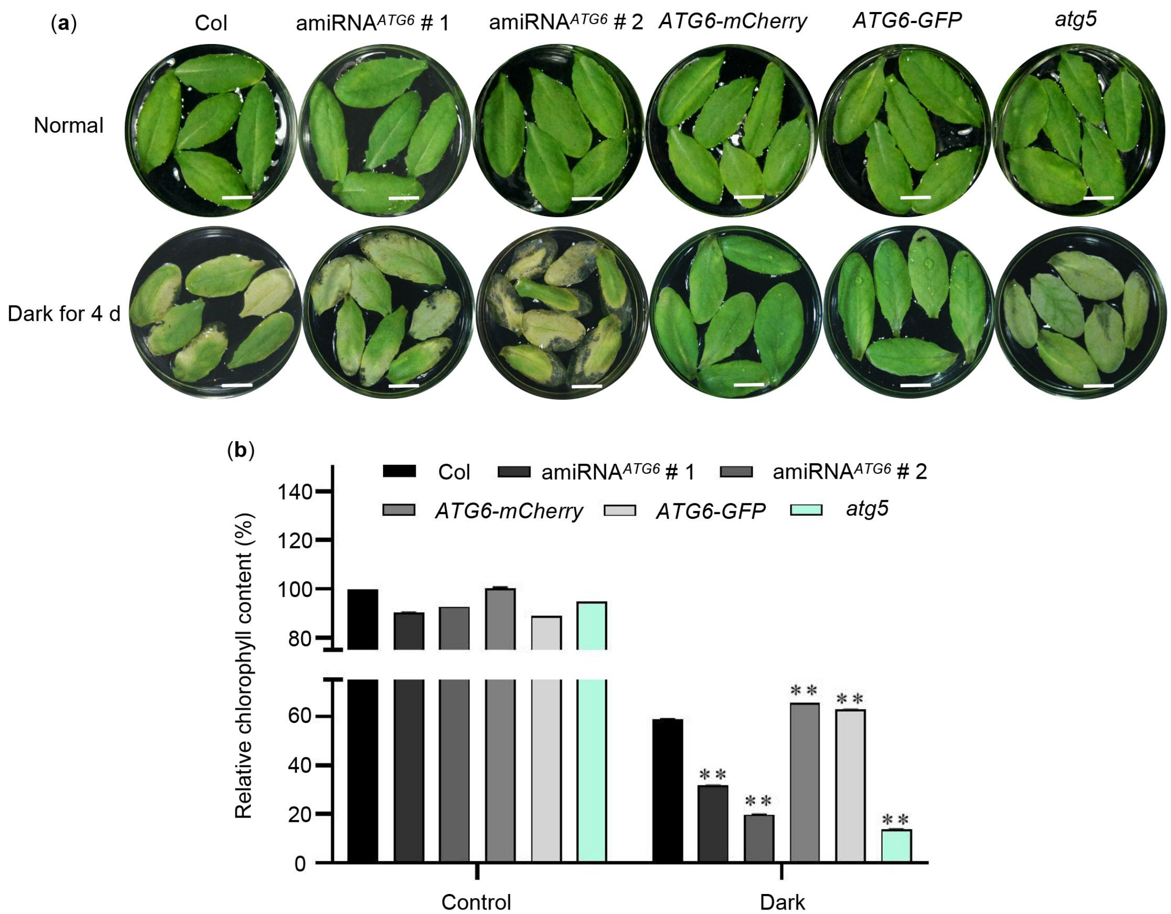


Fig. S8 Overexpression of *ATG6* delayed dark-induced leaf senescence. (a). Phenotypes of detached rosette leaves from 3-week-old of Col, amiRNA^{ATG6} # 1, amiRNA^{ATG6} # 2, *ATG6-mCherry*, *ATG6-GFP* and *atg5* under constant dark treatment for 4 days. Bar = 1 cm. (b). Relative chlorophyll content in (a). ** indicates that the significant difference compared to the Col is at the level of 0.01 (Student t test p-value, ** $p < 0.01$). Experiment was performed with three biological replicates.

Supplemental Figure 8

Supplemental Figure 9

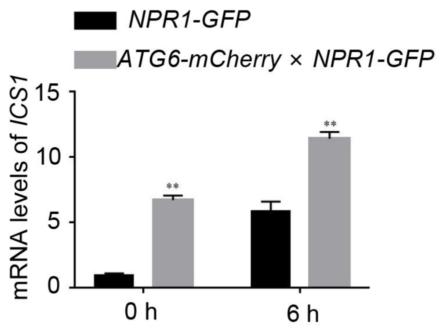


Fig. S9 Expression of *ICS1* under normal and 0.5 mM SA treatment conditions. Values are means ± SD (n = 3 biological replicates). The *AtActin* gene was used as the internal control.** indicates that the significant difference compared to the control is at the level of 0.01 (Student t test p value, ** p< 0.01). Experiment was performed with three biological replicates.

Supplemental Figure 10

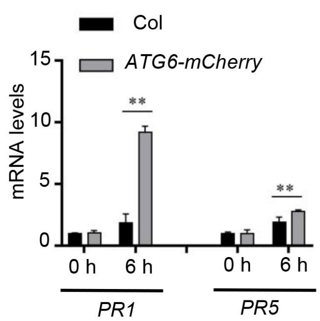


Fig. S10 Expression of *PR1* and *PR5* in Col and ATG6-mCherry under normal and *Pst* DC3000/*avrRps4* treatment. Values are means ± SD (n = 3 biological replicates). The *AtActin* gene was used as the internal control.** indicates that the significant difference compared to the control is at the level of 0.01 (Student t test p value, ** p< 0.01). Experiment was performed with three biological replicates.

Supplemental Figure 11

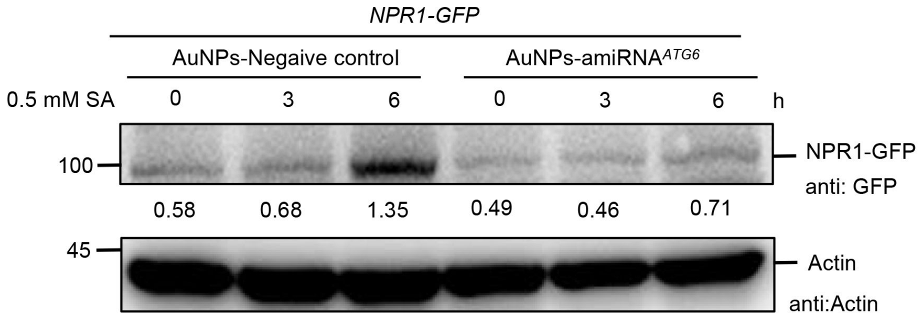


Fig. S11 The protein level of NPR1-GFP in *NPR1-GFP*/silencing *ATG6* and *NPR1-GFP*/Negative control. The mixture of AuNPs-amiRNA^{ATG6} or AuNPs-amiRNA$^{\text{Ngeative control}}$ was performed by pressure infiltration through the abaxial leaf surface. After 3 days, NPR1-GFP protein levels were detected by treating with 0.5 mM SA for 0, 3, and 6 h. Experiment was performed with three biological replicates.

Supplemental Figure 12

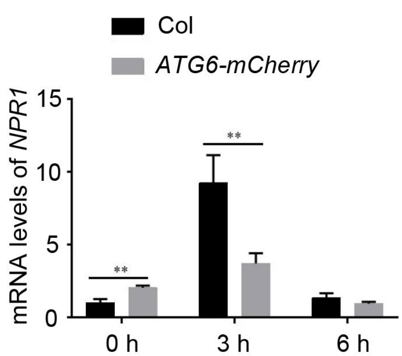


Fig. S12 Expression of *NPR1* in Col and *ATG6-mCherry* under normal and *Pst* DC3000/ *avrRps4* treatment. Values are means ± SD (n = 3 biological replicates). The *AtActin* gene was used as the internal control.** indicates that the significant difference compared to the control is at the level of 0.01 (Student t test p value, ** p< 0.01). Experiment was performed with three biological replicates.

Supplemental Figure 13

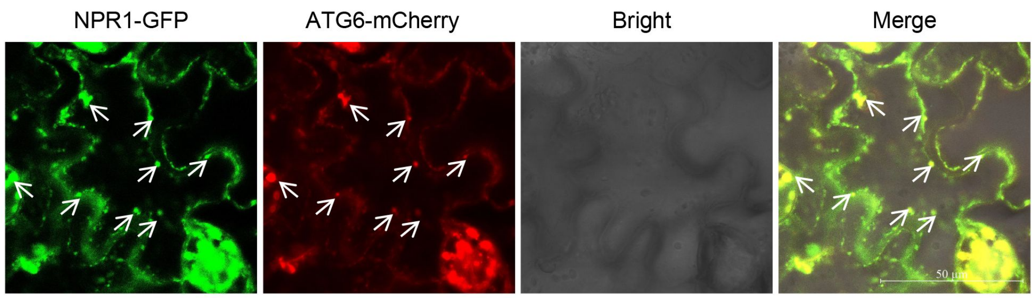
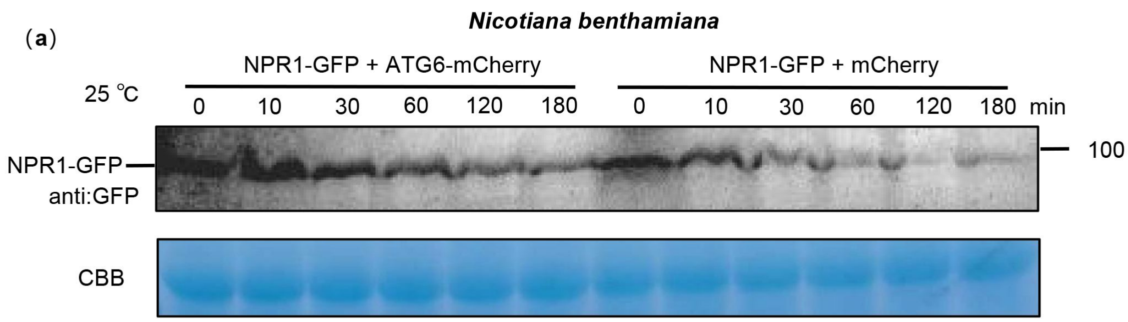


Fig. S13 Partial co-localization of ATG6-mCherry and SINCs-like condensates. ATG6-mCherry + NPR1-GFP were co-expressed in *N. benthamiana*. After 2 days, leaves were treated with 1 mM SA for 24 h. scale bar = 50 μm. Experiment was performed with three biological replicates.

Supplemental Figure 14



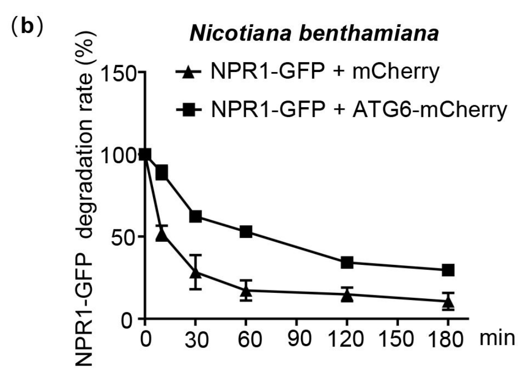


Fig S14. ATG6 improves the protein stability of NPR1 in *N. benthamiana*. (a). NPR1-GFP degradation assay in *N. benthamiana*. Total proteins from *N. benthamiana* co-transfected with mCherry + NPR1-GFP and ATG6-mCherry + NPR1-GFP were extracted. CBB was used as a control. **(b)**. Quantification of NPR1-GFP degradation rates in **(a)** using Image J. All experiments were performed with three biological replicates.

Supplemental Figure 15

148
NATINVLTRAFDIARTQTQVEQPLCLECMRVLSDKLEKEVEDVTRDVEAYEACVQRLEGETQDVLSEADFLKEK
KKIEEEERKLVAAIEETEKQNAEVNHQLKELEFKGNRFNELEDRYWQEFNNFQFQLIAHQEERDAILAKIEVSQ
295

Fig. S15 Structural analysis of acidic activation domains (AADs) in ATG6. Acidic (red) and hydrophobic (blue) amino acid residues in AADs.

Supplemental Figure 16

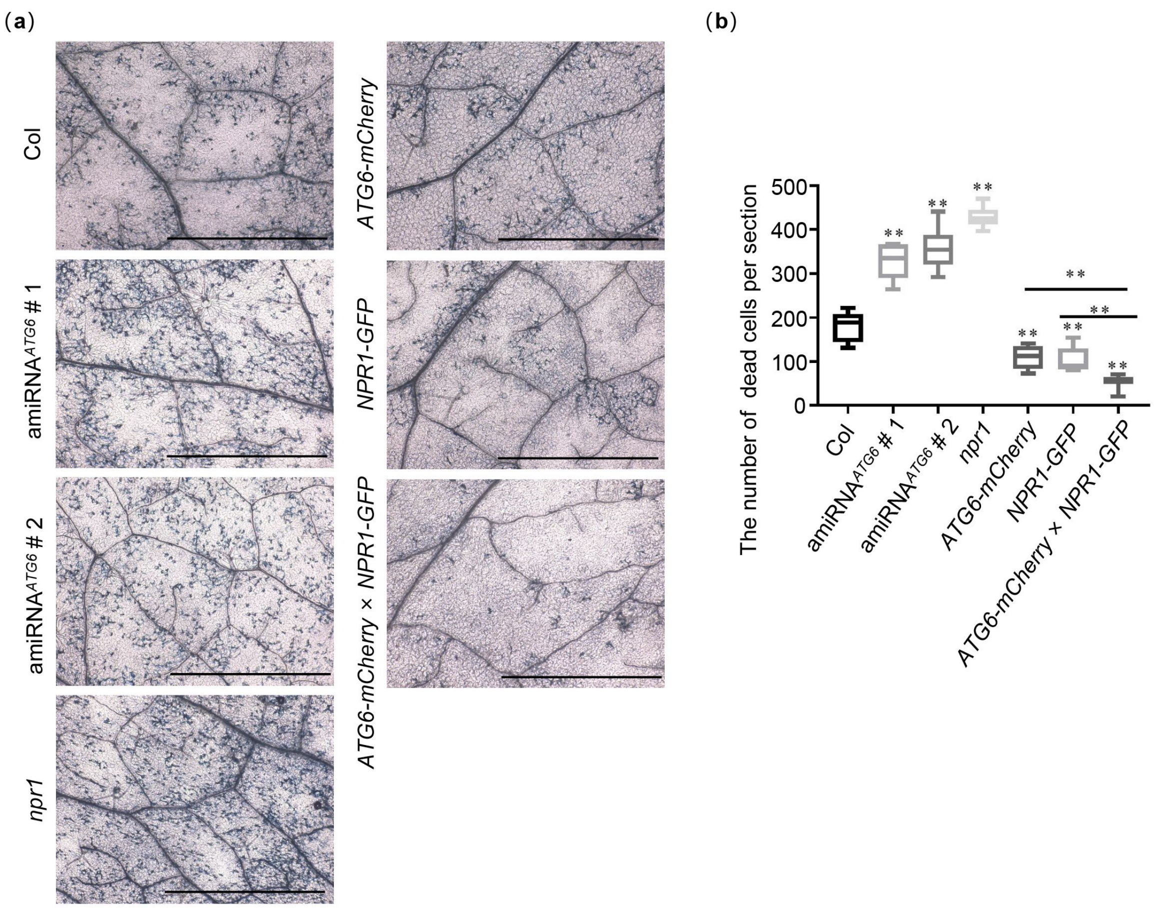


Fig. S16 ATG6 and NPR1 jointly inhibit *Pst* DC3000/*avrRps4*-induced cell dead . (a). Trypan blue staining showing cell death in the leaves of Col, amiRNA^{ATG6} # 1, amiRNA^{ATG6} # 2, *npr1*, *NPR1-GFP*, *ATG6-mCherry* and *ATG6-mCherry* × *NPR1-GFP*. A low dose of *Pst* DC3000/*avrRps4* (OD600 = 0.001) was infiltrated. After 3 days, Trypan blue staining was performed, Bar, 0.2 cm. (b). Number of dead cells per section in the leaves of Col, amiRNA^{ATG6} # 1, amiRNA^{ATG6} # 2, *npr1*, *NPR1-GFP*, *ATG6-mCherry* and *ATG6-mCherry* × *NPR1-GFP* in (a). ** indicates that the significant difference compared to the control is at the level of 0.01 (Student t test p value, ** $p < 0.01$). Experiment was performed with three biological replicates.

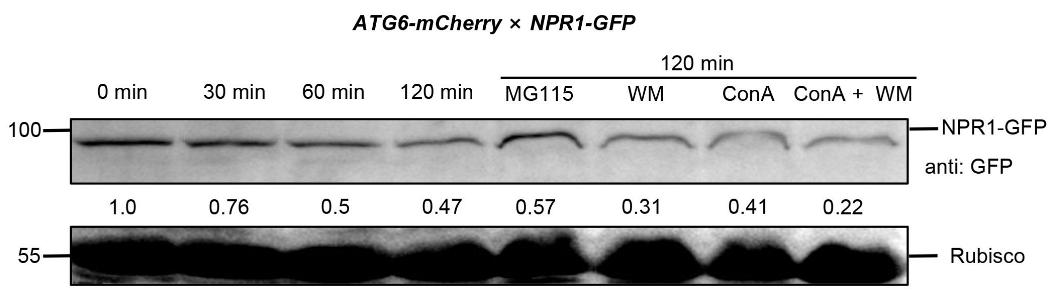


Fig. S17 NPR1-GFP degradation assay in *ATG6-mCherry* x *NPR1-GFP* *Arabidopsis*. Total proteins from 7-day-old seedlings of *ATG6-mCherry* x *NPR1-GFP* were extracted. These extracts were then incubated at room temperature (25℃) for 0~120 minutes to analyze the degradation rate of NPR1-GFP. To inhibit the proteasome pathway, 100 μM MG115 was utilized. Additionally, autophagy was inhibited using 5 μM concanamycin A and 30 μM Wortmannin. The analysis of NPR1-GFP was quantitatively performed using Image J software, and the corresponding numerical values were determined. Experiment was performed with two biological replicates.

Supplemental Figure 17

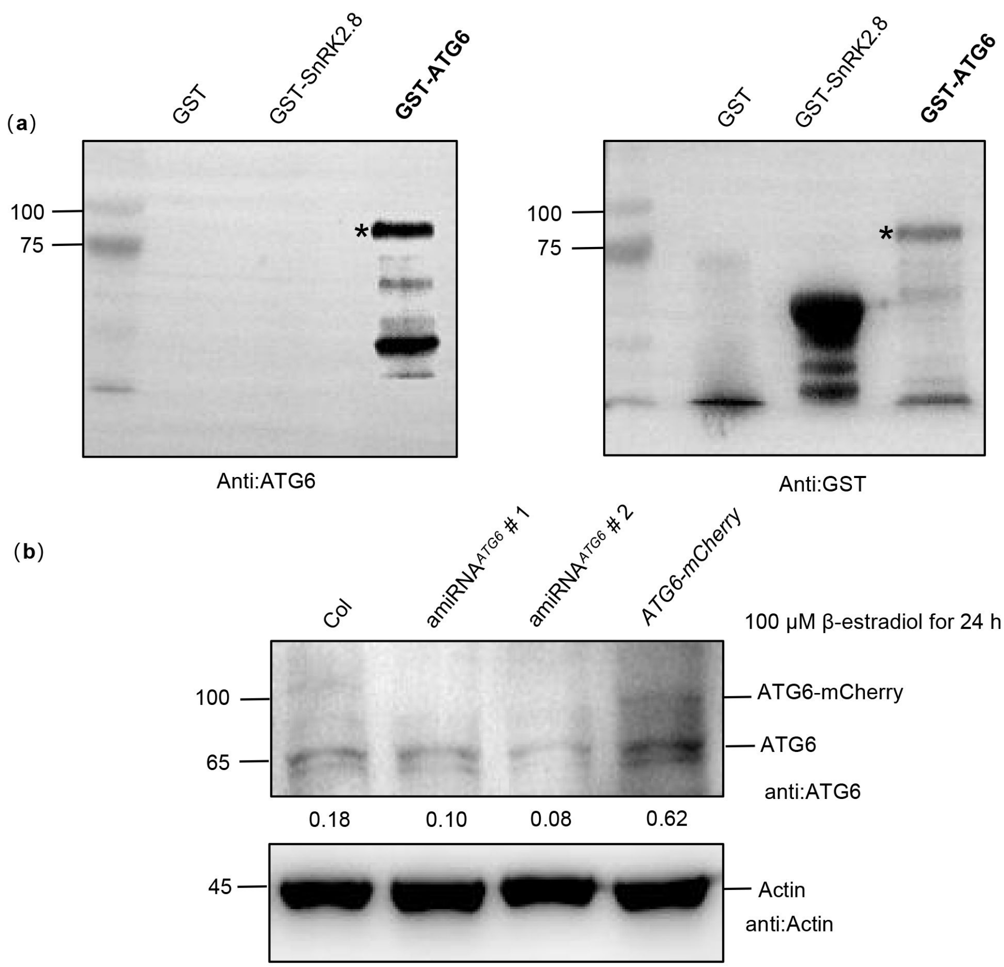


Fig. S18. Verification of ATG6 antibody specificity. (a). Prokaryotic expression of the GST-ATG6 fusion protein was used to verify the specificity of the ATG6 antibody, and GST and GST-SnRK2.8 were used as negative controls. GST-ATG6 bands was marked with a black asterisk. (b). The levels of ATG6 protein. Levels of ATG6-mCherry and endogenous ATG6 in Col, amiRNA^{ATG6} # 1, amiRNA^{ATG6} # 2 and *ATG6-mCherry* were detected after 100 μM estradiol treatment for 24 h. All experiments were performed with three biological replicates.

Supplemental Figure 18

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Editors

Reviewing Editor
Thabiso Motaung
University of Pretoria, Pretoria, South Africa

Senior Editor
Jürgen Kleine-Vehn
University of Freiburg, Freiburg, Germany

Reviewer #1 (Public Review):

The authors showed that autophagy-related genes are involved in plant immunity by regulating the protein level of the salicylic acid receptor, NPR1.

The experiments are carefully designed and the data is convincing. The authors did a good job of understanding the relationship between ATG6 and NRP1.

Comments on latest version:

The authors have already addressed all my comments. I have no further issues with the manuscript.

https://doi.org/10.7554/eLife.97206.3.sa2

Reviewer #2 (Public Review):

The manuscript by Zhang et al. explores the effect of autophagy regulator ATG6 on NPR1-mediated immunity. The authors propose that ATG6 directly interacts with NPR1 in the nucleus to increase its stability and promote NPR1-dependent immune gene expression and pathogen resistance. This novel role of ATG6 is proposed to be independent of its role in autophagy in the cytoplasm. The authors demonstrate through biochemical analysis that ATG6 interacts with NPR1 in yeast and very weakly in vitro. They further demonstrate using overexpression transgenic plants that in the presence of ATG6-mcherry the stability of NPR1-GFP and its nuclear pool is increased.

Comments on latest version:

The term "invasion" has to be replaced with infection, as it doesn't have much meaning to this particular study. I already explained this point in the first review, but authors did not address it throughout the manuscript.

In fig. 1e there's no statistical analysis. How can one show measurements from multiple samples without statistical analysis? All the data points have to be shown in the graph and statistics performed. In the arg6-npr1 and snrk-npr1 pairs no nuclear marker is included. How can one know where the nucleus is, particularly in such poor quality low res. images? The nucleus marker has to be included in this analysis and shown. This is an important aspect of the study as nuclear localization of ATG6 is proposed to be essential for its new function. Co-localization provided in the fig. S2 cannot complement this analysis, particularly since no cytoplasmic fraction is present for NPR1-GFP in fig. S2.

In the alignment in fig 2c, it is not explained what are the species the atg6 is taken from. The predicted NLS has to be shown in the context of either the entire protein sequence alignment or at least individual domain alignment with the indication of conserved residues (consensus). They have to include more species in the analysis, instead of including 3 proteins from a single species. Also, the predicted NLS in atg6 doesn't really have the classical type architecture, which might be an indication that it is a weak NLS, consistent with the fact that the protein has significant cytoplasmic accumulation. They also need to provide the NLS prediction cut-off score, as this parameter is a measure of NLS strength.

Line 150: the NLS sequence "FLKEKKKKK" is a wrong sequence.

In fig. 3d no explanation for the error bars is included, and what type of statistical analysis is performed is not explained.

https://doi.org/10.7554/eLife.97206.3.sa1

Author response:

The following is the authors' response to the previous reviews.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public Review):***
>
> *The authors showed that autophagy-related genes are involved in plant immunity by regulating the protein level of the salicylic acid receptor, NPR1.*
> *The experiments are carefully designed and the data is convincing. The authors did a good job of understanding the relationship between ATG6 and NRP1.*
>
> *The authors have addressed most of my previous concerns.*

Thank you so much for acknowledging our research. It is incredibly rewarding to see our work recognized. We hope that our findings will inspire new perspectives and foster further exploration in this area.

> ***Reviewer #2 (Public Review):***
>
> *The manuscript by Zhang et al. explores the effect of autophagy regulator ATG6 on NPR1-mediated immunity. The authors propose that ATG6 directly interacts with NPR1 in the nucleus to increase its stability and promote NPR1-dependent immune gene expression and pathogen resistance. This novel role of ATG6 is proposed to be independent of its role in autophagy in the cytoplasm. The authors demonstrate through biochemical analysis that ATG6 interacts with NPR1 in yeast and very weakly in vitro. They further demonstrate using overexpression transgenic plants that in the presence of ATG6-mcherry the stability of NPR1-GFP and its nuclear pool is increased.*
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> *Comments on revised version:*
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> *The authors demonstrate the correlation between overexertion of atg6 and higher stability and activity of npr1. They claim a novel activity of atg6 in the nucleus.*
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> *Overall, the experimental scope of the study is solid, however, the over-interpretation of the results substantially reduces the significance and value of this study for the target plant immunity readership.*

Thank you very much for you constructive and insightful comments, as well as for acknowledging the experimental scope of this study. In addition, we have made every effort to address the over-interpretation of the results, as per your comments, ensuring they are more accurate and concise. In the revised version, the modified content has been highlighted in blue to clearly indicate the changes made.

> ***Recommendations for the authors:***
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> ***Reviewer #1 (Recommendations For The Authors):***
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> *The authors have addressed most of my concerns. I have no further comments.*

Thank you so much for acknowledging our research. It is incredibly rewarding to see our work recognized. We hope that our findings will inspire new perspectives and foster further exploration in this area.

> ***Reviewer #2 (Recommendations For The Authors):***
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> *As I previously commented, in fig. 2a and c, the discrepancy between levels of atg6-mcherry in microscope image vs WB has to be explained. The explanation provided by the authors is incomplete and may mislead. The most likely reason for the difference is that the fluorescence signal in fig. 2a is predominantly from free mCherry, rather than the atg6-mcherry fusion. This has to be included in the main text to avoid misleading the reader.*

Thank you very much for you constructive and insightful comments, in response to your comments, we have incorporated the necessary explanations into the revised manuscript (lines 160-164).

> *In fig. 1B, the PD fraction has to show the size range of free GST. Also, please use "anti" to indicate that these are immunoblots,.*

Thank you for pointing this out. In the revised manuscript, we identified the range of free GST and used "anti:GST and anti:His" to indicate that these are immunoblots.

> *In fig 1C, the WB has to show the free GFP band in the input and IP fractions together with NPR1, rather than in separate blots.*

Thank you for bringing this to our attention. Fig. 1c has been replaced, and the updated image now shows the free GFP band in the input and IP fractions together with NPR1-GFP.

> *In fig. 1d, the bifc signal has to be quantified from multiple images across the biological repeats. Also, there's no significance in showing the chlorophyll autofluorescence. What is the purpose of this? They need to use a nuclear marker instead.*

Thank you for your suggestion. Based on your input, we utilized ImageJ software to quantify the YFP fluorescence signal. A total of n = 15 independent images were analyzed, and the corresponding results have been added to Figure 1e. Monitoring chlorophyll autofluorescence serves as a useful background signal, aiding in the distinction between the fluorescence signal of the target protein and background noise. This approach helps reduce potential signal overlap or interference during the experiment, thereby enhancing the reliability of the results.

> *Please provide a sequence alignment with multiple ATGs to show the conservation of the presumed bipartite NLS. This information has to be included in the main data.*

Thank you very much for your constructive and insightful comments. We analyzed the putative nuclear localization signal (NLS) in the ATG6 protein sequence using the online INSP (Identification of Nuclear Signal Peptide) prediction software (http://www.csbio.sjtu.edu.cn/bioinf/INSP/). The prediction results indicated the presence of a potential nuclear localization sequence "FLKEKKKKK" within the ATG6 protein, spanning from the 217th to the 223rd amino acid. Additionally, we utilized INSP to investigate the nuclear localization sequences of various ATG proteins (TaATG6a [1], TaATG6b [1], TaATG6c [1], SlATG8h [2]) that have been previously reported to localize in the nucleus. This analysis revealed a relatively conserved NLS sequence motif: "E/K-K/E-K-K-L/K-K" in these ATG proteins. In line with your suggestion, the results of this sequence comparison have been incorporated into the revised manuscript as Figure 2c. The revised manuscript includes a description of the corresponding results. (lines 146-156).

> *Fig. 3d and f, how many blots are used for this quantification? Please include all the individual analyzed blots in the supplementary data. In addition, if you present such quantification with error bars, then statistical analysis is required.*

Thank you for pointing this out. In Figure 3d, three independent blots were utilized for this quantification. In Figure 3f, two independent blots were used. The individual analyzed blots have been included in the supplementary Figure 7. We also conducted a statistical analysis as shown in Fig 3d and f, with a detailed description included in the legend section (lines 858 and 861).

> *In fig. 4, please indicate what is the normalizing gene. Also, what are the error bars?*

Thank you for pointing this out. In Fig.4, values are means ± SD (n = 3 biological replicates). The *AtActin* gene was used as the internal control. We have included a detailed description in the figure notes

> *In fig. 4b the labeling is missing.*

Thank you for bringing this to our attention. We have included the labeling for Fig. 4 in the revised manuscript.

> *Lines 236-239: this statement contradicts the data in fig. 5b: the levels of NPR1-GFP are actually reduced in the presence of atg6 at 24h. So, this result has to be described more accurately by stating that the increase is transient, and it is evident more at 8h, but not at 20-24h.*

Thank you very much for you constructive and insightful comments. We have revised the description of this section to provide a more accurate account of the results (lines 253-258).

Reference

(1) Yue J, Sun H, Zhang W, et al. Wheat homologs of yeast ATG6 function in autophagy and are implicated in powdery mildew immunity. BMC Plant Biol. 2015;15:95.

(2) Li F, Zhang M, Zhang C, et al. Nuclear autophagy degrades a geminivirus nuclear protein to restrict viral infection in solanaceous plants. New Phytol. 2020;225:1746-1761.

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